Theoretical Aspects of Modern Cryptography

Homework 1

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Problem 1 (Uniformity).

(i) Check that $(\{0,1\}^n, \oplus)$, where $n \in \mathbb{N}$ and $\oplus$ denotes the bitwise exclusive-or operation on $\{0,1\}^n$, forms a group.

Let U and V be random variables over G. Note that the group operation $\cdot$ induces the random variable $\cdot(U, V)$ over G, which will be denoted by $U \cdot V$. Suppose that U is uniformly random.

(ii) Suppose that U and V are independent. Prove that the random variable $U \cdot V$ is uniformly random.

(iii) Show that U and V may not be independent even if both V and $U \cdot V$ are uniformly random. (Hint: $(G, \cdot) = (\mathbb{Z}_3, +)$ and $V = U$.)

Answer (i).

- (Associativity): For any bits $b_1, b_2, b_3 \in \{0,1\}$, by enumerating all possible values of b_1, b_2, b_3, we see that $(b_1 \oplus b_2) \oplus b_3 = b_1 \oplus (b_2 \oplus b_3)$. Therefore, for any $a, b, c \in \{0,1\}^n$, $(a \oplus b) \oplus c = a \oplus (b \oplus c)$.
- (Identity): For any $a \in \{0,1\}^n$, $a \oplus 0^n = 0^n \oplus a = a$. Thus, 0^n is an (or rather, *the*) identity.
- (Inverse): For any $a \in \{0,1\}^n$, $a \oplus a = 0^n$, which is the identity of $(\{0,1\}^n, \oplus)$. That is, a is an (or rather, *the*) inverse of itself, for all $a \in \{0,1\}^n$.

Hence $(\{0,1\}^n, \oplus)$ forms a group. ⊛

Answer (ii).

Lemma 1.1. For any $a \in G$, $f_a : G \to G$ defined by $x \mapsto x \cdot a$ is a bijection.

Proof of lemma. Since f_a has inverse defined by $f_a^{-1} : x \mapsto x \cdot a^{-1}$ (one can check by definition that $f_a^{-1} \circ f_a = f_a \circ f_a^{-1} = \mathrm{id}_G$), f_a (and also f_a^{-1}) is bijective. ∎

Lemma 1.2. For any $a \in G$, $f_{-1} : G \to G$ defined by $x \mapsto x^{-1}$ is a bijection.

Proof of lemma. This is immediate since f_{-1} is the inverse of itself, implying f_{-1} is bijective. ∎

We calculate that for every $w \in G$

$$\begin{aligned}
p_{U \cdot V}(w) &= \sum_{u,v \in G,\ uv = w} p_{U,V}(u, v) \\
&= \sum_{u \in G} p_{U,V}(u, u^{-1}w) \\
&= \sum_{u \in G} p_U(u) p_V(u^{-1}w) && \text{(since } U, V \text{ are independent)} \\
&= \frac{1}{|G|} \sum_{u \in G} p_V(u^{-1}w) && \text{(since } U \text{ is uniformly distributed)} \\
&= \frac{1}{|G|} \sum_{x \in G} p_V(x) && \text{(by Lemma 1.1 and Lemma 1.2, } (f_w \circ f_{-1}) : u \mapsto u^{-1}w \text{ is a bijection)} \\
&= \frac{1}{|G|} \cdot 1 = \frac{1}{|G|},
\end{aligned}$$

from which it follows that $U \cdot V$ is also uniformly distributed. ⊛

Answer (iii). A counterexample is given by taking $(G, \cdot) \triangleq (\mathbb{Z}_3, +)$ and $U = V$, where U is uniformly distributed. First, U and V is clearly not independent since, for example,

$$p_{U,V}(0,1) = 0 \neq \frac{1}{3} \cdot \frac{1}{3} = p_U(0)p_V(1)$$

and thus $p_{U,V}(x,y) \neq p_U(x)p_V(y)$. However, since $U + V = U + U = 2U$ and U is uniformly distributed,

$$p_{U+V}(0) = p_{2U}(0) = p_U(0) = \frac{1}{3}, \quad p_{U+V}(1) = p_{2U}(1) = p_U(2) = \frac{1}{3}, \quad p_{U+V}(2) = p_{2U}(2) = p_U(1) = \frac{1}{3},$$

$U + V$ is also uniformly distributed. ⊛

Answer (iv). (Note that in $\mathbb{Z}_2$, $+$ is equivalent to $\oplus$.) Since $U \cdot V$ and U are uniformly distributed,

$$\frac{1}{2} = p_{U+V}(0) = p_{U,V}(0,0) + p_{U,V}(1,1) \tag{1.1}$$
$$\frac{1}{2} = p_{U+V}(1) = p_{U,V}(0,1) + p_{U,V}(1,0) \tag{1.2}$$
$$\frac{1}{2} = p_U(0) = p_{U,V}(0,0) + p_{U,V}(0,1) \tag{1.3}$$
$$\frac{1}{2} = p_U(1) = p_{U,V}(1,0) + p_{U,V}(1,1) \tag{1.4}$$

Therefore,

$$\begin{aligned}(1.1) - (1.3) \implies p_{U,V}(0,1) = p_{U,V}(1,1) &\implies p_V(1) = p_{U,V}(0,1) + p_{U,V}(1,1) = 2p_{U,V}(0,1) = 2p_{U,V}(1,1) \\ &\implies p_{U,V}(u,1) = \frac{1}{2}p_V(1) = p_U(u)p_V(1) \text{ for every } u \in \mathbb{Z}_2\end{aligned} \tag{1.5}$$

$$\begin{aligned}(1.2) - (1.3) \implies p_{U,V}(0,0) = p_{U,V}(1,0) &\implies p_V(0) = p_{U,V}(0,0) + p_{U,V}(1,0) = 2p_{U,V}(0,0) = p_{U,V}(1,0) \\ &\implies p_{U,V}(u,0) = \frac{1}{2}p_V(0) = p_U(u)p_V(0) \text{ for every } u \in \mathbb{Z}_2.\end{aligned} \tag{1.6}$$

Combining (1.5) and (1.6), we conclude that $p_{U,V}(u,v) = p_U(u)p_V(v)$ for all $u, v \in \mathbb{Z}_2$, which by definition means that U and V are independent. ⊛

Problem 2 (Markov).

Let $\mathcal{A}$ be a probabilistic algorithm with n bit input such that

$$\Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}(x) = 1] \geq p.$$

Prove that for any $0 < \varepsilon < p$, we have the size of the *good set* $|G_\varepsilon| \geq \varepsilon 2^n$, where

$$G_\varepsilon \triangleq \left\{ x \in \{0,1\}^n \;\middle|\; \Pr\big[\mathcal{A}(x) = 1\big] \geq p - \varepsilon \right\}.$$

You should be able to prove this claim via simple probabilistic operations.

Answer. Suppose to the contrary that $|G_\varepsilon| < \varepsilon 2^n$, equivalently,

$$\Pr_{x \leftarrow \{0,1\}^n} [x \in G_\varepsilon] = \frac{|G_\varepsilon|}{2^n} < \varepsilon.$$

Then, we deduce

$$\begin{aligned}
p &\leq \Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}(x) = 1] \\
&= \Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}(x) = 1 \wedge x \in G_\varepsilon] + \Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}(x) = 1 \wedge x \notin G_\varepsilon] \qquad \text{(since both events are disjoint)} \\
&= \Pr_{x \leftarrow \{0,1\}^n} [x \in G_\varepsilon] \Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}(x) = 1 \mid x \in G_\varepsilon] + \left(1 - \Pr_{x \leftarrow \{0,1\}^n} [x \in G_\varepsilon]\right) \Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}(x) = 1 \mid x \notin G_\varepsilon] \\
&< \varepsilon \cdot 1 + 1 \cdot (p - \varepsilon) \qquad \text{(by our assumption and the definition of } G_\varepsilon\text{)} \\
&= p,
\end{aligned}$$

which is a contradiction. Hence, it must be that $|G_\varepsilon| \geq \varepsilon 2^n$.

(The moral is: if the proportion of $x \in \{0,1\}^n$ such that $\mathcal{A}(x)$ wins with sufficiently high probability (at least $p - \varepsilon$) were too small (less than an ε fraction of $\{0,1\}^n$), then the average success probability $\Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}(x) = 1]$ could never reach p.) ⊛

Problem 3 (Fun with PRGs). For each of the following functions G, prove the statement: "if G_1 is a pseudorandom generator, then G is a pseudorandom generator" or provide a counterexample to demonstrate that it is not (assuming that PRGs exist).

(i) $G(x) = G_1(x) \oplus (x \parallel 0^{l(n)-n})$, where $|x| = n$.

(ii) $G(x) = G_1(x) \parallel G_1(x)$.

Answer (i). We show G is not always a PRG by providing a counterexample, i.e., a PRG G_1 such that the corresponding G is not a PRG, assuming that PRGs exist.

Let $H : \{0,1\}^* \to \{0,1\}^*$ be an arbitrary PRG where $|H(x)| = m(|x|)$ for some polynomial m. We define our $G_1 : \{0,1\}^* \to \{0,1\}^*$ as

$$G_1(x) \triangleq x_{[1]} \parallel H(x_{[2:n]}) \text{ where } n \text{ denotes } |x|,$$

and thus the corresponding $G : \{0,1\}^* \to \{0,1\}^*$ is

$$\begin{aligned} G(x) &\triangleq G_1(x) \oplus (x \parallel 0^{1+m(n-1)-n}) \\ &= (x_{[1]} \parallel H(x_{[2:n]})) \oplus (x_{[1]} \parallel x_{[2:n]} \parallel 0^{1+m(n-1)-n}) \\ &= 0 \parallel (H(x_{[2:n]}) \oplus (x_{[2:n]} \parallel 0^{1+m(n-1)-n})) \text{ where } n \text{ denotes } |x|. \end{aligned} \tag{3.1}$$

We claim that G_1 is a PRG whereas G is not.

Claim 3.1. G_1 is a PRG.

Proof of claim. It's trivial that, since H is a PRG, it is efficiently computable and expanding, and so does G_1. Hence it remains to show that $\{G_1(U_n)\}_n$ is pseudorandom.

Suppose to the contrary that $\{G_1(U_n)\}_n$ is not pseudorandom, that is, that there exists a PPT distinguisher D such that

$$\varepsilon(n) \triangleq \left|\Pr[D(G_1(U_n)) = 1] - \Pr[D(U_{1+m(n-1)}) = 1]\right|$$

is non-negligible (as a function of n). Equivalently, in the following security game (Game 1), the distinguisher D wins with probability $\frac{1}{2} + \frac{\varepsilon(n)}{2}$ for all $n \geq 1$, where ε is non-negligible.

$$\begin{array}{ccc}
\text{Ch} & & D \\
b \leftarrow \{0,1\} & & \\
s \leftarrow \begin{cases} G_1(U_n) = U_1 \parallel H(U_{n-1}), & b = 0 \\ U_{1+m(n-1)}, & b = 1 \end{cases} & & \\
 & \xrightarrow{\quad s \quad} & \\
 & \xleftarrow{\quad b' \quad} & \\
D \text{ wins if } b' = b & &
\end{array}$$

Figure 1: Game 1

Observe that the first bit of $G_1(U_n) = U_1 \parallel H(U_{n-1})$ and $U_{1+m(n-1)}$ distribute identically. Thus, Game 1 could be rewritten in the following equivalent game, denoted Game $1'$, wherein D still wins with the same probability $\frac{1}{2} + \frac{\varepsilon(n)}{2}$ for all $n \geq 1$.

$$\begin{array}{ccc}
\text{Ch} & & D \\
b \leftarrow \{0,1\} & & \\
c \leftarrow \{0,1\}, s \leftarrow \begin{cases} H(U_{n-1}), & b = 0 \\ U_{m(n-1)}, & b = 1 \end{cases} & & \\
s' := c \parallel s & & \\
 & \xrightarrow{\quad s' \quad} & \\
 & \xleftarrow{\quad b' \quad} & \\
D \text{ wins if } b' = b & &
\end{array}$$

Figure 2: Game $1'$

Formally speaking, for all $n \geq 1$,

$$\Pr[D \text{ wins Game } 1'] \equiv \Pr\left[b \leftarrow \{0,1\}, s' \leftarrow U_1 \parallel \begin{Bmatrix} H(U_{n-1}), & b = 0 \\ U_{m(n-1)}, & b = 1 \end{Bmatrix} : D(s') = b\right] = \frac{1}{2} + \frac{\varepsilon(n)}{2}. \tag{3.2}$$

From this, we provide a PPT distinguisher (a reduction) R that efficiently distinguishes between $\{H(U_n)\}_n$ and $\{U_{m(n)}\}_n$ with non-negligible advantage, i.e., wins the distinguishing game of $\{H(U_n)\}_n$ and $\{U_{m(n)}\}_n$ (Game 2) with non-negligible advantage. This implies $\{H(U_n)\}_n$ is not pseudorandom and thus H is not a PRG, contradicting to the assumption that H is a PRG.

■

Proof of claim, cont'd. We construct R as shown in the following game (Game 2). Namely, on input $s \in \{0,1\}^{m(n)}$, R uniformly samples $c \leftarrow \{0,1\}$, passes $c \parallel s$ to D, receives b from D, and finally outputs b. Our intuition is to have the distribution of what R passes to D ($c \parallel s$ in this case) be the same as that of what D takes as input (i.e., $s' = c \parallel s$) in Game 1.

$$
\begin{array}{ccccc}
\text{Ch} & & R & & D \\
b \leftarrow \{0,1\} & & & & \\
s \leftarrow \begin{cases} H(U_n), & b=0 \\ U_{m(n)}, & b=1 \end{cases} & & & & \\
 & \xrightarrow{\quad s \quad} & & & \\
 & & c \leftarrow \{0,1\} & & \\
 & & & \xrightarrow{\quad c\|s \quad} & \\
 & & & \xleftarrow{\quad b' \quad} & \\
 & \xleftarrow{\quad b' \quad} & & & \\
 & R \text{ wins if } b' = b & & &
\end{array}
$$

Figure 3: Game 2

It suffices to show that R runs in PPT and wins Game 2 with high probability. First, it's clear that R runs in PPT as D also runs in PPT. To see the latter holds, observe that for all $n \in \mathbb{N}$,

$$
\begin{aligned}
\Pr[R \text{ wins Game 2}] \equiv & \Pr\left[b \leftarrow \{0,1\}, s \leftarrow \begin{Bmatrix} H(U_n), & b=0 \\ U_{m(n)}, & b=1 \end{Bmatrix} : R(s) = b \right] \\
= & \Pr\left[b \leftarrow \{0,1\}, s \leftarrow \begin{Bmatrix} H(U_n), & b=0 \\ U_{m(n)}, & b=1 \end{Bmatrix}, c \leftarrow \{0,1\} : D(c \parallel s) = b \right] \\
= & \Pr\left[b \leftarrow \{0,1\}, s' \leftarrow U_1 \parallel \begin{Bmatrix} H(U_n), & b=0 \\ U_{m(n)}, & b=1 \end{Bmatrix} : D(s') = b \right] \quad (\text{by } s' \triangleq c \parallel s) \\
= & \frac{1}{2} + \frac{\varepsilon(n+1)}{2}, \quad (\text{by } (3.2))
\end{aligned}
$$

where $\varepsilon(n+1)$ is non-negligible.

Therefore, as reasoned above, the existence of the PPT distinguisher R implies by definition that $\{H(U_n)\}_n$ and $\{U_{m(n)}\}_n$ are not computationally indistinguishable, and thus H is not a PRG, contradicting the assumption that H is a PRG. ■

Claim 3.2. G is not a PRG given the G_1 defined above.

Proof of claim. To see this, we provide a PPT distinguisher D' that efficiently distinguishes between $\{G(U_n)\}_n$ and $\{U_{1+m(n-1)}\}_n$ with non-negligible advantage, i.e., wins the distinguishing game of $\{G(U_n)\}_n$ and $\{U_{1+m(n-1)}\}_n$ (Game 3) with non-negligible advantage. This implies $\{G(U_n)\}_n$ is not pseudorandom and thus G is not a PRG, finishing our proof.

We construct D' as shown in the following game (Game 3). Namely, on input $s \in \{0,1\}^n$, D' outputs 0 if $s_{[1]} = 0$ and 1 otherwise.

$$
\begin{array}{ccc}
\text{Ch} & & D' \\
b \leftarrow \{0,1\} & & \\
s \leftarrow \begin{cases} G(U_n), & b = 0 \\ U_{1+m(n-1)}, & b = 1 \end{cases} & & \\
& \xrightarrow{\quad s \quad} & \\
& & b' := s_{[1]} \\
& \xleftarrow{\quad b' \quad} & \\
D' \text{ wins if } b' = b & &
\end{array}
$$

Figure 4: Game 3

It suffices to show that D' runs in PPT and wins Game 3 with high probability. For the former, it's trivial that D' runs in PPT. To see the latter holds, observe that for all $n \geq 1$,

$$
\begin{aligned}
\Pr[D' \text{ wins Game 3}] &\equiv \Pr\left[b \leftarrow \{0,1\}, s \leftarrow \left\{ \begin{matrix} G(U_n), & b = 0 \\ U_{1+m(n-1)}, & b = 1 \end{matrix} \right\} : D(s) = b \right] \\
&= \Pr\left[b \leftarrow \{0,1\}, s \leftarrow \left\{ \begin{matrix} G(U_n), & b = 0 \\ U_{1+m(n-1)}, & b = 1 \end{matrix} \right\} : s_{[1]} = b \right] \\
&= \Pr[b \leftarrow \{0,1\} : b = 0] \Pr\left[b \leftarrow \{0,1\}, s \leftarrow \left\{ \begin{matrix} G(U_n), & b = 0 \\ U_{1+m(n-1)}, & b = 1 \end{matrix} \right\} : s_{[1]} = b \mid b = 0 \right] \\
&\quad + \Pr[b \leftarrow \{0,1\} : b = 1] \Pr\left[b \leftarrow \{0,1\}, s \leftarrow \left\{ \begin{matrix} G(U_n), & b = 0 \\ U_{1+m(n-1)}, & b = 1 \end{matrix} \right\} : s_{[1]} = b \mid b = 1 \right] \\
&= \Pr[b \leftarrow \{0,1\} : b = 0] \Pr\left[b \leftarrow \{0,1\}, x \leftarrow U_n : G(x)_{[1]} = 0 \mid b = 0 \right] \\
&\quad + \Pr[b \leftarrow \{0,1\} : b = 1] \Pr\left[b \leftarrow \{0,1\} : U_1 = 1 \mid b = 1 \right] \\
&= \Pr[b \leftarrow \{0,1\} : b = 0] \Pr\left[b \leftarrow \{0,1\}, x \leftarrow U_n : 0 = 0 \mid b = 0 \right] \\
&\quad + \Pr[b \leftarrow \{0,1\} : b = 1] \Pr\left[b \leftarrow \{0,1\} : U_1 = 1 \mid b = 1 \right] \\
&\quad \text{(since we know from (3.3) that the first bit of } G(x) \text{ is } 0) \\
&= \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} = \frac{1}{2} + \frac{\mu(n)}{2}
\end{aligned}
$$

where $\mu(n) \triangleq \frac{1}{2}$ is non-negligible.

Therefore, as reasoned above, the existence of the PPT distinguisher D' implies by definition that $\{G(U_n)\}_n$ and $\{U_{1+m(n-1)}\}_n$ are not computationally indistinguishable, and thus G is not a PRG. ■

Now we've completed the proof. Yay! ⊛

Answer (ii). We show G is not always a PRG by providing a counterexample, i.e., a PRG G_1 such that the corresponding G is not a PRG, assuming that PRGs exist.

Let $H : \{0,1\}^* \to \{0,1\}^*$ be an arbitrary PRG where $|H(x)| = m(|x|)$ for some polynomial m. We define our $G_1 : \{0,1\}^* \to \{0,1\}^*$ as

$$G_1(x) \triangleq x_{[1]} \parallel H(x_{[2:n]}) \text{ where } n \text{ denotes } |x|,$$

and thus the corresponding $G : \{0,1\}^* \to \{0,1\}^*$ is

$$\begin{aligned} G(x) &\triangleq G_1(\overline{x}) \parallel G_1(x) \\ &= \overline{x_{[1]}} \parallel H(\overline{x_{[2:n]}}) \parallel x_{[1]} \parallel H(x_{[2:n]}) \text{ where } n \text{ denotes } |x|. \end{aligned} \tag{3.3}$$

Observation 3.3. Note that for any $n \geq 1$ and $x \in \{0,1\}^n$, by (3.3), the 1st and $(1+m(n-1)+1)$th bit of $G(x)$ are always different, that is,

$$\begin{aligned} &G(x)_{[1]} \oplus G(x)_{[1+m(n-1)+1]} \\ =&(\overline{x_{[1]}} \parallel H(\overline{x_{[2:n]}}) \parallel x_{[1]} \parallel H(x_{[2:n]}))_{[1]} \oplus (\overline{x_{[1]}} \parallel H(\overline{x_{[2:n]}}) \parallel x_{[1]} \parallel H(x_{[2:n]}))_{[1+m(n-1)1]} \\ =&\overline{x_{[1]}} \oplus x_{[1]} = 1. \end{aligned}$$

We claim that G_1 is a PRG whereas G is not.

Claim 3.4. G_1 is a PRG.

Proof of claim. This is exactly Claim 3.1, which has already been proven. ■

Claim 3.5. G is not a PRG given the G_1 defined above.

Proof of claim. To see this, we provide a PPT distinguisher D' that efficiently distinguishes between $\{G(U_n)\}_n$ and $\{U_{2+2m(n-1)}\}_n$ with non-negligible advantage, i.e., wins the distinguishing game of $\{G(U_n)\}_n$ and $\{U_{2+2m(n-1)}\}_n$ (Game 4) with non-negligible advantage. This implies $\{G(U_n)\}_n$ is not pseudorandom and thus G is not a PRG, finishing our proof.

We construct D' as shown in the following game (Game 4). Namely, on input $s \in \{0,1\}^n$, D' outputs 0 if the 1st and $(1+m(n-1)+1)$th bit of s are different (equivalently, $s_{[1]} \oplus s_{[1+m(n-1)+1]} = 1$), and 1 otherwise. Our intuition comes from Observation 3.3.

$$
\begin{array}{ccc}
\text{Ch} & & D' \\
b \leftarrow \{0,1\} & & \\
s \leftarrow \begin{cases} G(U_n), & b = 0 \\ U_{2+2m(n-1)}, & b = 1 \end{cases} & & \\
 & \xrightarrow{\quad s \quad} & \\
 & & b' := \overline{s_{[1]} \oplus s_{[1+m(n-1)+1]}} \\
 & \xleftarrow{\quad b' \quad} & \\
 & D' \text{ wins if } b' = b &
\end{array}
$$

Figure 5: Game 4

It suffices to show that D' runs in PPT and wins Game 4 with high probability. For the former, it's trivial that D' runs in PPT. To see the latter holds, observe that for all $n \geq 1$,

$$
\begin{aligned}
&\Pr[D' \text{ wins Game } 4] \\
\equiv& \Pr\left[b \leftarrow \{0,1\}, s \leftarrow \begin{Bmatrix} G(U_n), & b = 0 \\ U_{2+2m(n-1)}, & b = 1 \end{Bmatrix} : D(s) = b \right] \\
=& \Pr\left[b \leftarrow \{0,1\}, s \leftarrow \begin{Bmatrix} G(U_n), & b = 0 \\ U_{2+2m(n-1)}, & b = 1 \end{Bmatrix} : s_{[1]} \oplus s_{[1+m(n-1)+1]} = \bar{b} \right] \\
=& \Pr[b \leftarrow \{0,1\} : b = 0] \Pr\left[b \leftarrow \{0,1\}, s \leftarrow \begin{Bmatrix} G(U_n), & b = 0 \\ U_{2+2m(n-1)}, & b = 1 \end{Bmatrix} : s_{[1]} \oplus s_{[1+m(n-1)+1]} = \bar{b} \mid b = 0 \right] \\
&\quad + \Pr[b \leftarrow \{0,1\} : b = 1] \Pr\left[b \leftarrow \{0,1\}, s \leftarrow \begin{Bmatrix} G(U_n), & b = 0 \\ U_{2+2m(n-1)}, & b = 1 \end{Bmatrix} : s_{[1]} \oplus s_{[1+m(n-1)+1]} = \bar{b} \mid b = 1 \right] \\
=& \Pr[b \leftarrow \{0,1\} : b = 0] \Pr\left[b \leftarrow \{0,1\}, x \leftarrow U_n : G(x)_{[1]} \oplus G(x)_{[1+m(n-1)+1]} = 1 \mid b = 0 \right] \\
&\quad + \Pr[b \leftarrow \{0,1\} : b = 1] \Pr\left[b \leftarrow \{0,1\}, s \leftarrow U_{2+2m(n-1)} : s_{[1]} \oplus s_{[1+m(n-1)+1]} = 0 \mid b = 1 \right] \\
=& \Pr[b \leftarrow \{0,1\} : b = 0] \Pr[1 = 1] + \Pr[b \leftarrow \{0,1\} : b = 1] \Pr[U_1 \oplus U_1 = 0] \quad \text{(by Observation 3.3)} \\
=& \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} = \frac{1}{2} + \frac{\mu(n)}{2}
\end{aligned}
$$

where $\mu(n) \triangleq \frac{1}{2}$ is non-negligible.

Therefore, the existence of the PPT distinguisher D' implies by definition that $\{G(U_n)\}_n$ and $\{U_{2+2m(n-1)}\}_n$ are not computationally indistinguishable, and thus G is not a PRG. ■

Now we've completed the proof. Yay! ⊛

Problem 4 (Fun with hardcore bits).

(i) Show that if a one-to-one function f has a hardcore predicate, then f is one-way.

(ii) Show that the assumption that f is one-to-one above is necessary. I.e., show that there exists a function f that is not one-to-one, has a hardcore predicate, but is not one-way.

Answer (i). Assume that the injection $f : \{0,1\}^* \to \{0,1\}^*$ has a hardcore predicate $h : \{0,1\}^* \to \{0,1\}$. To prove f is OW, first note that f is efficiently computable (i.e., can be computed in polynomial time). Then it remains to show that it's impossible for f to be efficiently invertible with non-negligible probability. To see this, suppose to the contrary that f is efficiently invertible with non-negligible probability, i.e., that there exists a PPT adversary $\mathbf{Adv}$ such that

$$\varepsilon(n) \triangleq \Pr_{x \leftarrow \{0,1\}^n} [f(\mathbf{Adv}(1^n, f(x))) = f(x)] \tag{4.1}$$

is non-negligible (as a function of n). Equivalently, in the following game (Game 1), the adversary $\mathbf{Adv}$ wins with probability $\varepsilon(n)$ for all $n \in \mathbb{N}$ where ε is non-negligible.

$$\begin{array}{ccc}
\text{Ch} & & \mathbf{Adv} \\
x \leftarrow \{0,1\}^n & & \\
y := f(x) & & \\
 & \xrightarrow{(1^n, y)} & \\
 & \xleftarrow{x'} & \\
\end{array}$$

$$\mathbf{Adv} \text{ wins if } f(x') = y$$

Figure 6: Game 1

From this, we provide a PPT predictor (a reduction) R that, for all $n \in \mathbb{N}$a and $x \leftarrow \{0,1\}^n$, efficiently predicts $h(x)$ given $f(x)$, i.e., wins the prediction game of $h(x)$ (Game 2), with high (non-negligible) probability. This implies h is not a hardcore predicate for f, contradicting to our assumption that h is.

We construct R as shown in the following game (Game 2). Namely, on input $(1^n, y := f(x))$, where $x \leftarrow \{0,1\}^n$, R passes $(1^n, y)$ to $\mathbf{Adv}$, receives x' from $\mathbf{Adv}$, checks if $f(x') = y$. If true, by the injectivity of f this means $x' = x$, so R outputs $h(x')$; otherwise, $x' \neq x$, so R simply outputs a uniformly random bit (i.e., makes a random guess). Our intuition is straightforward: since $\mathbf{Adv}$ efficiently finds a string in the preimage of $f(x)$ with non-negligible probability and f is injective, we have R exploit $\mathbf{Adv}$ to invert $f(x)$ back to x and then compute $h(x)$ itself. However, doing just this does not guarantee us a winning probability of at least $\frac{1}{2}$ that is required to break the hardcoreness of h. This is because even though $\mathbf{Adv}$ succeeds in inverting $f(x)$ with non-negligible probability $\varepsilon(n)$, it still fails with probability $1 - \varepsilon(n)$, and thus fails most of the time if $\varepsilon(n)$ is small; moreover, an incorrect x' output by $\mathbf{Adv}$ may lead to an incorrect $h(x')$. Knowing that R can check, by the injectivity of f, whether $\mathbf{Adv}$ has successfully inverted $f(x)$ back to x, we **let R randomly guess the**

bit $h(x)$ whenever it detects that $\mathbf{Adv}$ has failed to do so.

Ch		R		$\mathbf{Adv}$
$x \leftarrow \{0,1\}^n$				
$y := f(x)$				
	$\xrightarrow{(1^n, y)}$			
			$\xrightarrow{(1^n, y)}$	
			$\xleftarrow{x'}$	
		$c \leftarrow \{0,1\}$		
		$b := \begin{cases} h(x'), & f(x') = y \\ c, & \text{otherwise} \end{cases}$		
	$\xleftarrow{b}$			
		R wins if $b = h(x)$		

Figure 7: Game 2

It suffices to show that (1) R runs in PPT and (2) wins Game 2 with non-negligible probability. First, it's clear that R runs in PPT as $\mathbf{Adv}$ runs in PPT and $h(x')$ can be efficiently computed (which is because h is a hardcore predicate for f). To see the latter holds, observe that for all $n \in \mathbb{N}$,

$$\begin{aligned}
&\Pr[R \text{ wins Game 2}] \\
\equiv &\Pr_{x \leftarrow \{0,1\}^n}[R(1^n, f(x)) = h(x)] \\
= &\Pr_{x \leftarrow \{0,1\}^n, c \leftarrow \{0,1\}}\left[\begin{Bmatrix} h(x'), & f(x') = f(x) \\ c, & f(x') \neq f(x) \end{Bmatrix} = h(x)\right] \\
= &\Pr_{x \leftarrow \{0,1\}^n, c \leftarrow \{0,1\}}[f(x') = f(x)] \Pr_{x \leftarrow \{0,1\}^n, c \leftarrow \{0,1\}}\left[\begin{Bmatrix} h(x'), & f(x') = f(x) \\ c, & f(x') \neq f(x) \end{Bmatrix} = h(x) \mid f(x') = f(x)\right] \\
&+ \Pr_{x \leftarrow \{0,1\}^n, c \leftarrow \{0,1\}}[f(x') \neq f(x)] \Pr_{x \leftarrow \{0,1\}^n, c \leftarrow \{0,1\}}\left[\begin{Bmatrix} h(x'), & f(x') = f(x) \\ c, & f(x') \neq f(x) \end{Bmatrix} = h(x) \mid f(x') \neq f(x)\right] \\
= &\Pr_{x \leftarrow \{0,1\}^n}[f(x') = f(x)] \Pr_{x \leftarrow \{0,1\}^n}[h(x') = h(x) \mid f(x') = f(x)] \\
&+ \Pr_{x \leftarrow \{0,1\}^n}[f(x') \neq f(x)] \Pr_{x \leftarrow \{0,1\}^n, c \leftarrow \{0,1\}}[c = h(x) \mid f(x') \neq f(x)] \\
= &\Pr_{x \leftarrow \{0,1\}^n}[f(x') = f(x)] \Pr_{x \leftarrow \{0,1\}^n}[h(x') = h(x) \mid f(x') = f(x)] \\
&+ \left(1 - \Pr_{x \leftarrow \{0,1\}^n}[f(x') = f(x)]\right) \Pr_{x \leftarrow \{0,1\}^n}[U_1 = h(x) \mid f(x') \neq f(x)] \\
= &\varepsilon(n) \cdot \Pr_{x \leftarrow \{0,1\}^n}[h(x') = h(x) \mid f(x') = f(x)] + (1 - \varepsilon(n)) \cdot \frac{1}{2} && \text{(by (4.1))} \\
\geq &\varepsilon(n) \cdot \Pr_{x \leftarrow \{0,1\}^n}[f(x') = f(x) \mid f(x') = f(x)] + (1 - \varepsilon(n)) \cdot \frac{1}{2} && (4.2) \\
= &\varepsilon(n) \cdot 1 + (1 - \varepsilon(n)) \cdot \frac{1}{2} = \frac{1}{2} + \frac{\varepsilon(n)}{2}
\end{aligned}$$

where we denote $x' \triangleq \mathbf{Adv}(1^n, f(x))$. Note that the inequality (4.2) is simply a consequence of the injectivity of f:

$$h(x') = h(x) \impliedby x' = x \overset{\text{injectivity of } f}{\iff} f(x') = f(x).$$

Since $\frac{\varepsilon(n)}{2}$ is non-negligible, R wins Game 2 with non-negligible probability.

Therefore, as reasoned above, the existence of the PPT predictor R implies by definition that h is not a hardcore predicate for f, contradicting the assumption that h is. ⊛

Answer (ii). Let $f : \{0,1\}^* \to \{0,1\}^*$ and $h : \{0,1\}^* \to \{0,1\}$ be defined by

$$f(x) \triangleq 0^n, \quad h(x) \triangleq x_{[1]},$$

where n denotes $|x|$.

It's immediate that f is not injective, so it suffices to show that (1) h is a hardcore predicate for f and that (2) f is not OW.

Claim 4.1. h is a hardcore predicate for f given the h and f defined above.

Proof of claim. To prove h is a hardcore predicate for f, first note that both h and f are efficiently computable (i.e., can be computed in polynomial time). Then it remains to show that, given $f(x)$, it's impossible for $h(x)$ to be efficiently predictable with non-negligible probability.

To see this, observe that for any PPT predictor $\mathbf{Adv}$ and all $n \in \mathbb{N}$,

$$h(U_n) = (U_n)_{[1]} = U_1 \tag{4.3}$$

and therefore we have

$$\begin{aligned}
&\Pr_{x \leftarrow \{0,1\}^n}[\mathbf{Adv}(1^n, f(x)) = h(x)] \\
=&\Pr_{x \leftarrow \{0,1\}^n}[\mathbf{Adv}(1^n, 0^n) = h(x)] \\
&\qquad\text{(by the definition of } f\text{)} \\
=&\Pr[\mathbf{Adv}(1^n, 0^n) = h(U_n)] \\
=&\Pr[\mathbf{Adv}(1^n, 0^n) = U_1] \\
&\qquad\text{(by (4.3))} \\
=&\Pr[\mathbf{Adv}(1^n, 0^n) = 1]\Pr[U_1 = 1 \mid \mathbf{Adv}(1^n, 0^n) = 1] + \Pr[\mathbf{Adv}(1^n, 0^n) = 0]\Pr[U_1 = 0 \mid \mathbf{Adv}(1^n, 0^n) = 0] \\
=&\Pr[\mathbf{Adv}(1^n, 0^n) = 1]\Pr[U_1 = 1] + \Pr[\mathbf{Adv}(1^n, 0^n) = 0]\Pr[U_1 = 0] \\
&\qquad\text{(since the random variables } U_1 \text{ and } \mathbf{Adv}(1^n, 0^n) \text{ are independent)} \\
=&\Pr[\mathbf{Adv}(1^n, 0^n) = 1] \cdot \frac{1}{2} + \Pr[\mathbf{Adv}(1^n, 0^n) = 0] \cdot \frac{1}{2} \\
=&\frac{1}{2} + 0,
\end{aligned}$$

where 0 is a negligible function of n. This means given $f(x)$, $h(x)$ is merely efficiently predictable with negligible probability.

Therefore, by definition, h is indeed a hardcore predicate for f. ■

Claim 4.2. f is not OW given the f defined above.

Proof of claim. To see this, we provide a PPT adversary $\mathbf{Adv}$ that efficiently inverts f, i.e., wins the inverting game of f (Game 3), with non-negligible probability, which, by definition, implies f is not OW.

We construct $\mathbf{Adv}$ as shown in the following game (Game 3). Namely, on input $(1^n, y := f(x))$, where $x \leftarrow \{0,1\}$, $\mathbf{Adv}$ always outputs 0^n. Our intuition is that since $f(x) = 0^n$ for all $x \in \{0,1\}^n$ and $n \in \mathbb{N}$, all strings of length n lie in the same preimage $f^{-1}(0^n)$, so any such string will do.

$$
\begin{array}{ccc}
\text{Ch} & & \mathbf{Adv} \\
x \leftarrow \{0,1\}^n & & \\
y := f(x) = 0^n & & \\
 & \xrightarrow{(1^n, y=0^n)} & \\
 & & x' := 0^n \\
 & \xleftarrow{x'} & \\
\end{array}
$$

$$\mathbf{Adv} \text{ wins if } f(x') = y$$

Figure 8: Game 3

It suffices to verify that $\mathbf{Adv}$ (1) runs in PPT and (2) wins Game 3 with non-negligible probability. Firstly, it's trivial that $\mathbf{Adv}$ runs in PPT. To see the latter holds, observe that for any $n \in \mathbb{N}$,

$$
\begin{aligned}
\Pr[\mathbf{Adv} \text{ wins Game 3}] \equiv & \Pr_{x \leftarrow \{0,1\}^n}[f(\mathbf{Adv}(1^n, f(x))) = f(x)] \\
= & \Pr_{x \leftarrow \{0,1\}^n}[0^n = 0^n] && \text{(by the definition of } f\text{)} \\
= & 1
\end{aligned}
$$

which is clear non-negligible (as a function of n).

Hence, by definition, f is not OW, as desired. ■

Now we've completed the proof. Yay! ⊛

Problem 5 (Recap on Hardness Amplification).

In this problem, you would be asked to review and extend the hardness amplification proof. We first give a brief recap to the outline of proof. In this proof, we aim to show that given a (t, ε)-one way function f, we can construct a (t', ε')-one-way function $f^{(k)}$, where

$$f^{(k)}(x_1, \ldots, x_k) = (f(x_1), \ldots, f(x_k)).$$

We showed that we can achieve an $\varepsilon' = ((1+\delta)\varepsilon)^k + \gamma$, which is almost exponentially smaller to ε.

The proof of the one-wayness of $f^{(k)}$, not surprisingly, requires reduction. In other words, we must start from an adversary $\mathbf{Adv}_k$ with at least ε' (which is smaller) winning rate on inverting $f^{(k)}$, and construct an reduction $\mathcal{R}$, using $\mathbf{Adv}_k$ as a subroutine, and achieve at least ε (which is larger) winning rate on inverting f.

The reduction R is sketched as follows:

1. $\mathcal{R}$ receives an challenge $y = f(x)$ for random $x \leftarrow \{0,1\}^n$ from the challenger.
2. For each $i \in [k]$:
   - Retry the following procedure for l times:
     (a) For each $j \neq i$, sample random $x_j \leftarrow \{0,1\}^n$ and compute $y_j = f(x_j)$. Also set $y_i = y$.
     (b) Run $\mathbf{Adv}_k$ with input $\vec{y} = (y_1, \ldots, y_k)$, obtain output $\vec{x} = (x_1, \ldots, x_k)$.
     (c) If $f(x_i) = y$, return x_i to the challenger.

In this problem, we will discuss the details on the proof showing that $\mathcal{R}$ indeed inverts f with probability at least ε.

(i) **(Tightness)** Show that given an $\mathbf{Adv}_k$ with win rate ε^k, no matter how large l is set, the win rate of the given reduction $\mathcal{R}$ cannot always be amplified to larger than ε. (**Hint.** You probably need to "design" an $\mathbf{Adv}_k$ with specific winning pattern.)

(ii) **(Achieving Strong One-Way)** In the lecture, we proved a concrete version of hardness amplification. Now try to apply the concrete result and get the asymptotic version. Prove that, given a $\varepsilon(n)$-weak one-way function f, where $1 - \varepsilon(n) \geq \frac{1}{p(n)}$ for some polynomial $p(n)$. You can choose a suitable k (probably as a function of n) such that $f^{(k)}$ is strong one-way. (**Hint.** As always, you should prove by contradiction. Try starting with an $\mathbf{Adv}_k$ with some non-negligible win rate $q(n)$ and get the parameters tuned. Note that you cannot directly set γ to negligible. (Why?))

(iii) **(Twisting the Reductions)** Let us try to change the way we choose i (the starred line in $\mathcal{R}$) in the reduction $\mathcal{R}$. Consider a reduction $\mathcal{R}_{\text{random}}$ which, instead of looping for each i, only sample one random $i \in \{1, \ldots, k\}$ to try for the l iterations. Does $\mathcal{R}_{\text{random}}$ give a similar amplification result?

Answer (i). Let S be an arbitrary subset of $\{0,1\}^n$ of size $\lceil \varepsilon 2^n \rceil$ (say, $S \triangleq \{(0)_2, (1)_2, \ldots, (\lceil \varepsilon 2^n \rceil - 1)_2\}$), and denote $(\cdot)_i$ by the ith entry of a vector. Assume that $\varepsilon 2^n \in \mathbb{N}$ and that for all $(x_1, \ldots, x_k) \in S^k$,

$$\Pr\left[\forall i \in [k].f\left(\mathbf{Adv}_k(1^{nk}, (f(x_1), \ldots, f(x_k)))_i\right) = f(x_i)\right] = \frac{\varepsilon^k 2^{nk}}{\lceil \varepsilon 2^n \rceil^k} \tag{5.1}$$

$$\Pr\left[\forall i \in [k].f\left(\mathbf{Adv}_k(1^{nk}, (f(x_1), \ldots, f(x_k)))_i\right) \neq f(x_i)\right] = 1 - \frac{\varepsilon^k 2^{nk}}{\lceil \varepsilon 2^n \rceil^k}, \tag{5.2}$$

and for all $(x_1, \ldots, x_k) \notin S^k$,

$$\Pr\left[\forall i \in [k]. f\left(\mathbf{Adv}_k(1^{nk}, (f(x_1), \ldots, f(x_k)))_i\right) \neq f(x_i)\right] = 1. \tag{5.3}$$

This assumption is valid because

$$0 < \frac{\varepsilon^k 2^{nk}}{\lceil \varepsilon 2^n \rceil^k} \leq \frac{\varepsilon^k 2^{nk}}{(\varepsilon 2^n)^k} = 1$$

and

$$\begin{aligned}
& \Pr[\mathbf{Adv}_k \text{ wins}] \\
\equiv & \Pr_{x_1,\ldots,x_k \leftarrow \{0,1\}^n}\left[f^{(k)}(\mathbf{Adv}_k(1^{nk}, (f(x_1), \ldots, f(x_k)))) = (f(x_1), \ldots, f(x_k))\right] \\
= & \Pr_{x_1,\ldots,x_k \leftarrow \{0,1\}^n}\left[\forall i \in [k]. f(\mathbf{Adv}_k(1^{nk}, (f(x_1), \ldots, f(x_k)))_i) = f(x_i)\right] \\
= & \Pr_{x_1,\ldots,x_k \leftarrow \{0,1\}^n}\left[(x_1, \ldots, x_k) \in S^k\right] \\
& \qquad \Pr_{x_1,\ldots,x_k \leftarrow \{0,1\}^n}\left[\forall i \in [k]. f(\mathbf{Adv}_k(1^{nk}, (f(x_1), \ldots, f(x_k)))_i) = f(x_i) \mid (x_1, \ldots, x_k) \in S^k\right] \\
& + \Pr_{x_1,\ldots,x_k \leftarrow \{0,1\}^n}\left[(x_1, \ldots, x_k) \notin S^k\right] \\
& \qquad \Pr_{x_1,\ldots,x_k \leftarrow \{0,1\}^n}\left[\forall i \in [k]. f(\mathbf{Adv}_k(1^{nk}, (f(x_1), \ldots, f(x_k)))_i) = f(x_i) \mid (x_1, \ldots, x_k) \notin S^k\right] \\
= & \Pr_{x_1,\ldots,x_k \leftarrow \{0,1\}^n}\left[(x_1, \ldots, x_k) \in S^k\right] \cdot \frac{\varepsilon^k 2^{nk}}{\lceil \varepsilon 2^n \rceil^k} + \Pr_{x_1,\ldots,x_k \leftarrow \{0,1\}^n}\left[(x_1, \ldots, x_k) \notin S^k\right] \cdot 0 \quad \text{(by assumptions (5.1) and(5.3))} \\
= & \frac{\lceil \varepsilon 2^n \rceil^k}{2^{nk}} \cdot \frac{\varepsilon^k 2^{nk}}{\lceil \varepsilon 2^n \rceil^k} + \left(1 - \frac{\lceil \varepsilon 2^n \rceil^k}{2^{nk}}\right) \cdot 0 \qquad \text{(by the definition of } S\text{)} \\
= & \varepsilon^k,
\end{aligned}$$

the latter of which is consistent with our assumption that $\mathbf{Adv}_k$ wins with probability ε^k.

Then we make an observation:

Observation 5.1.

$$\begin{aligned}
\mathbf{Adv}_k \text{ correctly inverts } f(x_i) \text{ for } \textbf{some } i \in [k] &\iff \mathbf{Adv}_k \text{ correctly inverts } f(x_i) \text{ for } \textbf{all } i \in [k] \\
&\implies \text{every } x_i \text{ is in } S.
\end{aligned}$$

That is,

$$\begin{aligned}
\exists i \in [k]. f(\mathbf{Adv}_k(1^{nk}, (f(x_1), \ldots, f(x_k)))_i) = f(x) &\iff \forall i \in [k]. f(\mathbf{Adv}_k(1^{nk}, (f(x_1), \ldots, f(x_k)))_i) = f(x) \\
&\implies (x_1, \ldots, x_k) \in S^k.
\end{aligned}$$

The winning rate of $\mathcal{R}$ is therefore upper bounded by

$$
\begin{aligned}
&\Pr[\mathcal{R}\text{ wins}]\\
\equiv&\Pr_{x\leftarrow\{0,1\}^n}\left[f(\mathcal{R}(1^n,f(x)))=f(x)\right]\\
=&\Pr_{x\leftarrow\{0,1\}^n,\ \forall i\in[k]\ \forall t\in l\ \forall j\in[k-1]\setminus[i].x_j^{(i,t)}\leftarrow\{0,1\}^n}\Bigg[\exists i\in[k]\ \exists t\in[l]\text{ s.t.}\\
&\quad f\left(\mathbf{Adv}_k\left(1^{nk},\left(f(x_1^{(i,t)}),\dots,\underbrace{f(x)}_{i\text{th entry}},\dots,f(x_k^{(i,t)})\right)\right)_i\right)=f(x)\text{ in the }(i,t)\text{th trial}\Bigg]\\
\le&\Pr_{x\leftarrow\{0,1\}^n,\ \forall i\in[k]\ \forall t\in l\ \forall j\in[k-1]\setminus[i].x_j^{(i,t)}\leftarrow\{0,1\}^n}\left[(f(x_1^{(i,t)}),\dots,\underbrace{f(x)}_{i\text{th entry}},\dots,f(x_k^{(i,t)}))\in S^k\right]\qquad(\text{by Observation 5.1})\\
\le&\Pr_{x\leftarrow\{0,1\}^n}\left[f(x)\in S\right]\\
=&\frac{\lceil\varepsilon 2^n\rceil}{2^n}\qquad(\text{by the definition of }S)\\
=&\frac{\varepsilon 2^n}{2^n}\qquad(\text{by the assumption that }\varepsilon 2^n\in\mathbb{N})\\
=&\varepsilon,
\end{aligned}
$$

as desired.

$\circledast$

Answer (ii). Let $k(n)\triangleq np(n)$. We prove that $f^{(k(n))}$ is strongly OW.

To show this, suppose to the contrary that $f^{(k(n))}$ is NOT strongly OW, i.e., there exists a PPT adversary $\mathbf{Adv}_k$ such that

$$
\begin{aligned}
q(n)\triangleq&\Pr_{x_1,\dots,x_{k(n)}\leftarrow\{0,1\}^n}[\mathbf{Adv}_k\text{ wins }\mathsf{G}_k]\\
\equiv&\Pr_{x_1,\dots,x_{k(n)}\leftarrow\{0,1\}^n}\left[f^{(k(n))}(\mathbf{Adv}_k(1^{nk(n)},(f(x_1),\dots,f(x_{k(n)}))))=(f(x_1),\dots,f(x_{k(n)}))\right]
\end{aligned}
$$

is non-negligible, where G_k is the following game.

$$
\begin{array}{ccc}
\text{Ch} & & \mathbf{Adv}_k\\
x_1,\dots,x_{k(n)}\leftarrow\{0,1\}^n & & \\
 & \xrightarrow{(1^{nk(n)},\vec{y}=f^{(k(n))}(x_1,\dots,x_{k(n)})=(f(x_1),\dots,f(x_{k(n)})))} & \\
 & \xleftarrow{\vec{x}'=(x_1',\dots,x_{k(n)}')} & \\
 & \mathbf{Adv}_k\text{ wins if }f(x_i')=f(x_i)\ \forall i\in[k(n)] &
\end{array}
$$

Figure 9: Game G_k

Since $q(n)$ is non-negligible, by definition, there exists $c>0$ such that

$$
q(n)\ge\frac{1}{n^c}\tag{5.4}
$$

for infinitely many $n\in\mathbb{N}$. In the following context, we consider only these n's, and show that we can construct a PPT reduction $\mathcal{R}$ that efficiently inverts f (i.e., wins the game G) with probability at least $\varepsilon(n)$ for these (infinitely many) $n \in \mathbb{N}$ that are sufficiently large. By definition, this breaks the $\varepsilon(n)$-weak one-wayness of f, contradicting to the assumption that f is $\varepsilon(n)$-weak OW.

We construct $\mathcal{R}$ as described in the following game G:

Ch | $\mathcal{R}$ | $\mathbf{Adv}_k$

$x \leftarrow \{0,1\}^n$

$y := f(x)$

$\xrightarrow{(1^n, y)}$

Repeat $\frac{k(n)}{\alpha(n)} \cdot n$ times $\left(l \in \left[\frac{k(n)}{\alpha(n)} \cdot n\right]\right)$:

For $i \in [k(n)]$:

$x_j^{(i,l)} \leftarrow \{0,1\}^n \; \forall j \in [k(n)] \setminus \{i\}$

$\xrightarrow{(1^{nk(n)}, (f(x_1^{(i,l)}), \ldots, \overbrace{y}^{i\text{th entry}}, \ldots, f(x_{k(n)}^{(i,l)})))}$

$\xleftarrow{(x_1'^{(i,l)}, \ldots, x_{k(n)}'^{(i,l)})}$

Set $x^\star := x_i'^{(i,l)}$ if $f(x_i'^{(i,l)}) = y$;

repeat the double loop otherwise.

$\xleftarrow{x^\star}$

$\mathcal{R}$ wins if $f(x^\star) = f(x)$

Figure 10: Game G

where $\alpha(n)$ denotes

$$\alpha(n) \triangleq \frac{1}{2n^c p(n)}. \tag{5.5}$$

For $i \in [k(n)]$, we define a *good* set G_i of *good* elements by

$$G_i \triangleq \left\{ x \in \{0,1\}^n \;\middle|\; \Pr_{x_1 \ldots, x_{k(n)} \leftarrow \{0,1\}^n} \left[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \mid x_i = x\right] \geq \frac{\alpha(n)}{k(n)} \right\}.$$

Before lower bounding the probability that $\mathcal{R}$ wins G, we claim two properties of the set G_i.

Claim 5.2. If $x \in G_i$ for some $i \in [k(n)]$, then

$$\Pr\left[\mathcal{R} \text{ wins } \mathsf{G}\right] \geq 1 - e^{-n}.$$

Proof of claim.

$$\begin{aligned}
\Pr\left[\mathcal{R} \text{ wins } \mathsf{G}\right] &\geq \Pr\left[\exists l \in \left[\frac{k(n)}{\alpha(n)} \cdot n\right].\mathcal{R} \text{ finds an answer in the } (i,l)\text{th trial}\right] \\
&\geq \Pr_{\forall l \in \left[\frac{k(n)}{\alpha(n)} \cdot n\right].\forall j \in [k(n)]\setminus\{i\}.x_j^{(i,l)} \leftarrow \{0,1\}^n}\left[\exists l \in \left[\frac{k(n)}{\alpha(n)} \cdot n\right].\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \text{ in the } (i,l)\text{th trial}\right] \\
&= 1 - \Pr_{\forall l \in \left[\frac{k(n)}{\alpha(n)} \cdot n\right].\forall j \in [k(n)]\setminus\{i\}.x_j^{(i,l)} \leftarrow \{0,1\}^n}\left[\forall l \in \left[\frac{k(n)}{\alpha(n)} \cdot n\right].\mathbf{Adv}_k \text{ loses } \mathsf{G}_k \text{ in the } (i,l)\text{th trial}\right] \\
&= 1 - \prod_{l \in \left[\frac{k(n)}{\alpha(n)} \cdot n\right]}\left(1 - \Pr_{\forall j \in [k(n)]\setminus\{i\}.x_j^{(i,l)} \leftarrow \{0,1\}^n}\left[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \text{ in the } (i,l)\text{th trial}\right]\right) \\
&= 1 - \left(1 - \Pr_{x_1,\ldots,x_{k(n)} \leftarrow \{0,1\}^n}\left[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \mid x_i = x\right]\right)^{\frac{k(n)}{\alpha(n)} \cdot n} \\
&\geq 1 - \left(1 - \frac{\alpha(n)}{k(n)}\right)^{\frac{k(n)}{\alpha(n)} \cdot n} \qquad \text{(by the definition of } G_i\text{)} \\
&\geq 1 - \left(\lim_{m \to \infty}\left(1 - \frac{1}{m}\right)^m\right)^n \qquad \left(\text{since } \left\{\left(1 - \frac{1}{n}\right)^n\right\}_{n \in \mathbb{N}} \text{ is an increasing sequence with limit at } \infty\right) \\
&= 1 - e^{-n} \qquad \left(\text{since } \lim_{m \to \infty}\left(1 - \frac{1}{m}\right)^m = e^{-1}\right)
\end{aligned}$$

■

Claim 5.3. For infinitely many $n \in \mathbb{N}$, there exists $i^* \in [k(n)]$ such that $\Pr_{x \leftarrow \{0,1\}^n}[x \in G_{i^*}] \geq (\frac{1}{n^c} - \alpha(n))^{\frac{1}{k(n)}}$.

Proof of claim. Let $n \in Nbb$ be such that (5.4) holds. Suppose to the contrary that $\Pr_{x \leftarrow \{0,1\}^n}[x \in G_i] < (\frac{1}{n^c} - \alpha(n))^{\frac{1}{k(n)}}$ for all $i \in [k(n)]$. Observe that

$$\begin{aligned}
&\Pr_{x_1,\ldots,x_{k(n)} \leftarrow \{0,1\}^n}\left[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \wedge (\exists i \in [k(n)].x_i \notin G_i)\right] \\
&= \Pr_{x_1,\ldots,x_{k(n)} \leftarrow \{0,1\}^n}\left[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \wedge \left(\bigvee_{i \in [k(n)]} x_i \notin G_i\right)\right] \\
&\leq \sum_{i \in [k(n)]} \Pr_{x_1,\ldots,x_{k(n)} \leftarrow \{0,1\}^n}\left[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \wedge x_i \notin G_i\right] \qquad \text{(by union bound)} \\
&\leq \sum_{i \in [k(n)]} \Pr_{x_1,\ldots,x_{k(n)} \leftarrow \{0,1\}^n}\left[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \mid x_i \notin G_i\right] \\
&< \sum_{i \in [k(n)]} \frac{\alpha(n)}{k(n)} = \alpha(n). \qquad \text{(by the definition of } G_i\text{)}
\end{aligned}$$

■

Proof of claim (cont'd). Observe also that

$$\begin{aligned}
\Pr_{x_1,\dots,x_{k(n)}\leftarrow\{0,1\}^n}[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \wedge (\forall i\in[k(n)].x_i\in G_i)] &\le \Pr_{x_1,\dots,x_{k(n)}\leftarrow\{0,1\}^n}[\forall i\in[k(n)].x_i\in G_i] \\
&= \Pr_{x_1,\dots,x_{k(n)}\leftarrow\{0,1\}^n}\left[\bigwedge_{i\in[k(n)]} x_i\in G_i\right] \\
&= \prod_{i\in[k(n)]} \Pr_{x_1,\dots,x_{k(n)}\leftarrow\{0,1\}^n}[x_i\in G_i] \\
&< \prod_{i\in[k(n)]} (\frac{1}{n^c}-\alpha(n))^{\frac{1}{k(n)}} \qquad \text{(by assumption)} \\
&= \frac{1}{n^c}-\alpha(n).
\end{aligned}$$

Combining both observations, we can upper bound the winning probability of $\mathbf{Adv}_k$ by $q(n)$:

$$\begin{aligned}
q(n) &\triangleq \Pr_{x_1,\dots,x_{k(n)}\leftarrow\{0,1\}^n}[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k] \\
&= \Pr_{x_1,\dots,x_{k(n)}\leftarrow\{0,1\}^n}[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \wedge (\exists i\in[k(n)].x_i\notin G_i)] \\
&\qquad + \Pr_{x_1,\dots,x_{k(n)}\leftarrow\{0,1\}^n}[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \wedge (\forall i\in[k(n)].x_i\in G_i)] \\
&< \alpha(n) + \left(\frac{1}{n^c}-\alpha(n)\right) \\
&= \frac{1}{n^c} \\
&\le q(n) \qquad \text{(by (5.4))},
\end{aligned}$$

which is a contradiction. ■

Moreover, we show several inequalities that come in handy later in our proof.

Observation 5.4. Since $\varepsilon(n)$ represents a probability, $p(n) \ge \frac{1}{1-\varepsilon(n)} \ge \frac{1}{1-0} = 1$.

Claim 5.5. There exists $N_1 \in \mathbb{N}$ such that $np(n) \ge 1 + 2c\ln(n)p(n)$ for all $n \ge N_1$.

Proof of claim. Since $\lim_{n\to\infty} \frac{1+2c\ln(n)}{n} = 0$, there exists $N_1 \in \mathbb{N}$ such that $\left|\frac{1+2c\ln(n)}{n} - 0\right| \le 1$ and thus $1 + 2c\ln(n) \le n$ for all $n \ge N_1$. Hence, for all $n \ge N_1$

$$np(n) \ge (1+2c\ln(n))p(n) = p(n) + 2c\ln(n)p(n) \overset{\text{by Observation 5.4}}{\ge} 1 + 2c\ln(n)p(n).$$

■

Claim 5.6. $-\ln\left(1-\frac{1}{2p(n)}\right) \ge \frac{1}{2p(n)}$

Proof of claim. Since $(1-\frac{1}{x})^x$ is increasing for $x \in [1,\infty)$ and $2p(n) \geq 2$ by Observation 5.4, we have

$$\left(1-\frac{1}{2p(n)}\right)^{2p(n)} \leq \lim_{x\to\infty}\left(1-\frac{1}{x}\right)^x = e^{-1}.$$

Taking $\ln(\cdot)$ on both sides, we deduce

$$\left(1-\frac{1}{2p(n)}\right)^{2p(n)} \leq e^{-1} \iff 2p(n)\ln\left(1-\frac{1}{p(n)}\right) \leq -1 \iff -\ln\left(1-\frac{1}{2p(n)}\right) \geq \frac{1}{2p(n)}.$$

■

Claim 5.7. There exists $N_2 \in \mathbb{N}$ such that $1-\frac{1}{2p(n)} \geq \frac{1-\frac{1}{p(n)}}{1-e^{-n}}$ for all $n \geq N_2$.

Proof of claim. Observe that

$$1-\frac{1}{2p(n)} \geq \frac{1-\frac{1}{p(n)}}{1-e^{-n}} \iff 1-e^{-n} \geq \frac{1-\frac{1}{p(n)}}{1-\frac{1}{2p(n)}} = 1-\frac{1}{2p(n)-1} \iff e^n \geq 2p(n)-1. \quad (5.6)$$

Since $p(n) = \mathsf{poly}(n)$, $\lim_{n\to\infty}\frac{e^n}{2p(n)-1} = \infty$, which implies there exists $N_2 \in \mathbb{N}$ such that $\left|\frac{e^n}{2p(n)-1}-0\right| \geq 1$ and thus $e^n \geq 2p(n)-1$ for all $n \geq N_2$. Therefore, by (5.6), $\left(1-\frac{1}{2p(n)}\right)(1-e^{-n}) \geq 1-\frac{1}{p(n)}$ for all $n \geq N_2$. ■

Combining these claims, we see that for $n \geq \max\{N_1, N_2\}$ such that (5.4) holds (of which there are infinitely many),

$$\begin{aligned}
\Pr_{x\leftarrow\{0,1\}^n}[\mathcal{R}\text{ wins }\mathsf{G}] &\geq \Pr_{x\leftarrow\{0,1\}^n}[\mathcal{R}\text{ wins }\mathsf{G} \wedge x \in G_{i^*}] \\
&= \Pr_{x\leftarrow\{0,1\}^n}[x \in G_{i^*}]\Pr_{x\leftarrow\{0,1\}^n}[\mathcal{R}\text{ wins }\mathsf{G} \mid x \in G_{i^*}] \\
&\geq \left(\frac{1}{n^c}-\alpha(n)\right)^{\frac{1}{k(n)}}(1-e^{-n}) && \text{(by Claim 5.2 and Claim 5.3)} \\
&\overset{\bigstar}{\geq} 1-\frac{1}{p(n)} \\
&\geq \varepsilon(n),
\end{aligned}$$

where the second-to-last inequality ($\bigstar$) is given by

$$
\begin{aligned}
& \left(\frac{1}{n^c} - \alpha(n)\right)^{\frac{1}{k(n)}} (1 - e^{-n}) \geq 1 - \frac{1}{p(n)} \\
\Longleftarrow\ & \left(\frac{1}{n^c} - \alpha(n)\right)^{\frac{1}{k(n)}} \geq 1 - \frac{1}{2p(n)} \overset{\text{by Claim 5.7}}{\geq} \frac{1 - \frac{1}{p(n)}}{1 - e^{-n}} \\
\Longleftarrow\ & \left(\frac{1}{n^c} - \alpha(n)\right)^{\frac{1}{k(n)}} \overset{\text{by def. of } \alpha(n)}{=} \left(\frac{1}{n^c}\left(1 - \frac{1}{2p(n)}\right)\right)^{\frac{1}{k(n)}} \geq 1 - \frac{1}{2p(n)} \\
\Longleftarrow\ & \frac{1}{n^c}\left(1 - \frac{1}{2p(n)}\right) \geq \left(1 - \frac{1}{2p(n)}\right)^{k(n)} \\
\Longleftarrow\ & -c\ln(n) + \ln\left(1 - \frac{1}{2p(n)}\right) \geq k(n)\ln\left(1 - \frac{1}{2p(n)}\right) && \text{(taking } \ln(\cdot) \text{ on both sides)} \\
\Longleftarrow\ & k(n) \geq 1 + \frac{c\ln(n)}{\frac{1}{2p(n)}} \overset{\text{by Claim 5.6}}{\geq} 1 + \frac{c\ln(n)}{-\ln\left(1 - \frac{1}{2p(n)}\right)} && \left(\text{since } \ln\left(1 - \frac{1}{2p(n)}\right) < 0\right) \\
\Longleftarrow\ & k(n) \overset{\text{by def. of } k(n)}{=} np(n) \geq 1 + 2c\ln(n)p(n) \\
\Longleftarrow\ & np(n) \overset{\text{by Claim 5.5}}{\geq} 1 + 2c\ln(n)p(n).
\end{aligned}
$$

Now it suffices to show that $\mathcal{R}$ runs in PPT. To see this, notice that the (maximum) running time of $\mathcal{R}$ is upper bounded by

$$k(n)\left(\frac{k(n)}{\alpha(n)} \cdot n\right)((k(n)-1)t_f(n) + t_{\mathbf{Adv}_k}(nk(n)) + t_f(n)) = 2n^{c+3}p^3(n)(np(n)t_f(n) + t_{\mathbf{Adv}_k}(n^2p(n))),$$

where $t_f(m)$ and $t_{\mathbf{Adv}_k}(m)$ denotes the computation time of f and running time of $\mathbf{Adv}_k$, respectively, on an input of length m. Note that $t_f(m)$ is a polynomial as f is $\varepsilon(m)$-weak OW, and that $t_{\mathbf{Adv}_k}(m)$ is a probabilistic polynomial of m because $\mathbf{Adv}_k$ runs in PPT. In addition, since $t_{\mathbf{Adv}_k}(m)$ and $n^2p(n)$ are both (probabilistic) polynomials (of m and n, respectively), $t_{\mathbf{Adv}_k}(n^2p(n))$ is a probabilistic polynomial of n. Therefore, since $p(n)$, $t_f(n)$ are polynomials and $t_{\mathbf{Adv}_k}(n^2p(n))$ is a probabilistic polynomial, the running time of $\mathcal{R}$ is a probabilistic polynomial as well, i.e., $\mathcal{R}$ runs in PPT.

Hence, $\mathcal{R}$ runs in PPT and inverts f (i.e., wins the game G) with probability at least $\varepsilon(n)$ for infinitely many $n \in \mathbb{N}$ (that are sufficiently large), f is not $\varepsilon(n)$-weak OW, contradicting the assumption that f is. This completes the proof, yay! ⊛

Answer (iii). In the proof of hardness amplification (concrete version), if we replace $\mathcal{R}$ with the following reduction $\mathcal{R}_{\mathsf{random}}$:

Ch		$\mathcal{R}_{\mathsf{random}}$		$\mathbf{Adv}_k$
$x \leftarrow \{0,1\}^n$ $y := f(x)$				
	$\xrightarrow{\quad y \quad}$			
		$i \leftarrow [k]$ Repeat $\frac{2kn}{\gamma}$ times $\left(l \in \left[\frac{2kn}{\gamma}\right]\right)$: $x_j \leftarrow \{0,1\}^n \ \forall j \in [k] \setminus \{i\}$		
			$\xrightarrow{(f(x_1^{(l)}),\ldots,\overbrace{y}^{i\text{th entry}},\ldots,f(x_k^{(l)}))}$	
			$\xleftarrow{(x_1'^{(l)},\ldots,x_k'^{(l)})}$	
		Set $x^\star := x_i'^{(l)}$ if $f(x_i'^{(l)}) = y$; repeat the loop otherwise.		
	$\xleftarrow{\quad x^\star \quad}$			
		$\mathcal{R}_{\mathsf{random}}$ wins if $f(x^\star) = f(x)$		

Figure 11: Game G

then the last step of our original proof will no longer work:

$$
\begin{aligned}
& \Pr_{x \leftarrow \{0,1\}^n}\left[\mathcal{R}_{\mathsf{random}} \text{ wins } \mathsf{G}\right] \\
\geq & \Pr_{x \leftarrow \{0,1\}^n, i \leftarrow [k]}\left[\mathcal{R}_{\mathsf{random},i} \text{ wins } \mathsf{G} \wedge x \in G_{i^*} \wedge i = i^*\right] \\
= & \Pr_{x \leftarrow \{0,1\}^n}[x \in G_{i^*}] \Pr_{i \leftarrow [k]}[i = i^*] \Pr_{x \leftarrow \{0,1\}^n}\left[\mathcal{R}_{\mathsf{random},i} \text{ wins } \mathsf{G} \;\middle|\; x \in G_{i^*} \wedge i = i^*\right] && \text{(by cond. prob. and independence)} \\
\geq & (1+\delta)\varepsilon \cdot \frac{1}{k} \cdot (1 - e^{-n}) && \text{(by Claim 5.8 and Claim 5.9)} \\
> & \frac{\varepsilon}{k} && \left(\text{since } \delta > \frac{e^{-n}}{1 - e^{-n}}\right) \\
\not\geq & \varepsilon,
\end{aligned}
$$

where $\mathcal{R}_{\mathsf{random},i}$ denotes $\mathcal{R}_{\mathsf{random}}$ in which $i \in [k]$ is sampled, and the aforementioned claims are the following.

Claim 5.8. If $x \in G_i$ for some $i \in [k]$, then

$$\Pr\left[\mathcal{R}_{\mathsf{random}} \text{ wins } \mathsf{G}\right] \geq 1 - e^{-n}.$$

Proof of claim. The argument is identical to the one for $\mathcal{R}$ in the original proof. ■

Claim 5.9. There exists $i^* \in [k]$ such that $\Pr_{x \leftarrow \{0,1\}^n}[x \in G_{i^*}] \geq (1+\delta)\varepsilon$.

Proof of claim. This is proven in the original proof. ■

As a side note here, recall that in the original proof we define a *good* set G_i as

$$G_i \triangleq \left\{x \in \{0,1\}^n \;\middle|\; \Pr_{x_1 \ldots, x_{k(n)} \leftarrow \{0,1\}^n}\left[\mathbf{Adv}_k \text{ wins } \mathsf{G}_k \mid x_i = x\right] \geq \frac{\gamma}{2k}\right\}.$$

Notice that the last step will not work because we couldn't guarantee $(1+\delta)(1-e^{-n})k^{-1} > 1$. Therefore, if we must use $\mathcal{R}_{\mathsf{random}}$ as the reduction, we can still prove a somewhat weaker theorem albeit the original proof does not work.

Theorem (Our modified theorem). Let $k \in \mathbb{N}$. If $f : \{0,1\}^n \to \{0,1\}^m$ is (t,ε)-OW, then the function $f^{(k)} : \{0,1\}^{n\times k} \to \{0,1\}^{m\times k}$ defined by $f^{(k)}(x_1,\dots,x_k) = (f(x_1),\dots,f(x_k))$ is (t',ε')-OW with

- $t' = \frac{\gamma}{4kn}t$
- $\varepsilon' = ((1+\delta)\varepsilon)^k + \gamma$ for $\delta > \frac{k}{1-e^{-n}} - 1$

provided that $t' \geq kt_f$, where t_f denotes the computation time of f.

Proof of the modified theorem. The proof of this theorem is essentially the same as that of hardness amplification (concrete version) proven in class, except for the running time analysis part and the last step, which shows the reduction inverts f with probability more than ε and thus yields a contradiction.

In the last step of our proof, we deduce:

$$\begin{aligned}
&\Pr_{x\leftarrow\{0,1\}^n}\left[\mathcal{R}_{\mathsf{random}} \text{ wins } \mathsf{G}\right] \\
\geq &\Pr_{x\leftarrow\{0,1\}^n, i\leftarrow[k]}\left[\mathcal{R}_{\mathsf{random},i} \text{ wins } \mathsf{G} \wedge x \in G_{i^*} \wedge i = i^*\right] \\
= &\Pr_{x\leftarrow\{0,1\}^n}[x\in G_{i^*}]\Pr_{i\leftarrow[k]}[i=i^*]\Pr_{x\leftarrow\{0,1\}^n}\left[\mathcal{R}_{\mathsf{random},i} \text{ wins } \mathsf{G} \,\middle|\, x\in G_{i^*}\wedge i = i^*\right] && \text{(by cond. prob. and independence)} \\
\geq &(1+\delta)\varepsilon\cdot\frac{1}{k}\cdot(1-e^{-n}) && \text{(by Claim 5.8 and Claim 5.9)} \\
> &\varepsilon. && \left(\text{since } \delta > \frac{k}{1-e^{-n}} - 1\right)
\end{aligned}$$

Moreover, $\mathcal{R}_{\mathsf{random}}$ runs in time t since

$$\frac{2kn}{\gamma}\cdot((k-1)t_f + t' + t_f) \leq \frac{2kn}{\gamma}\cdot 2t' = \frac{2kn}{\gamma}\cdot 2\cdot\frac{\gamma}{4kn}t = t.$$

We then conclude that $\mathcal{R}_{\mathsf{random}}$ breaks the (t,ε)-one-wayness of f, which is a contradiction. Thus the proof is completed. ■

⊛

Problem 6 (Expansion of a PRG).

Let $G : \{0,1\}^n \to \{0,1\}^{n+1}$ be a PRG. For any polynomial $l(\cdot)$, define $G' : \{0,1\}^n \to \{0,1\}^{l(n)}$ as follows:

$$\begin{aligned} G'(s) &\triangleq b_1 \dots b_{l(n)}, \text{ where} \\ x_0 &:= s \\ x_{i+1} \,\|\, b_{i+1} &:= G(x_i),\ i = 0, 1, \dots, l(n) - 1. \end{aligned}$$

Then G' is a PRG.

Answer. It's trivial that G' is efficiently computable (in deterministic polynomial time) and expanding (since we may assume that $|G'(s)| = l(|s|) > |s|$). Thus, it remains to show that $\{G'(U_n)\}_n$ is pseudorandom, which we prove by contradiction.

Suppose to the contrary that $\{G'(U_n)\}_n$ is not pseudorandom, which means $\{G'(U_n)\}_n$ and $\{U_{l(n)}\}_n$ are not computationally indistinguishable. That is, that there exists a PPT distinguisher D such that

$$\varepsilon(n) \triangleq \left|\Pr[D(G'(U_n)) = 1] - \Pr[D(U_{l(n)}) = 1]\right| \tag{6.1}$$

is non-negligible. Equivalently, in the following security game G', the distinguisher D wins with probability $\frac{1}{2} + \frac{\varepsilon(n)}{2}$ for all $n \in \mathbb{N}$, where ε is non-negligible.

$$\begin{array}{ccc} \text{Ch} & & D \\ b \leftarrow \{0,1\} & & \\ y \leftarrow \begin{cases} G'(U_n), & b = 0 \\ U_{l(n)}, & b = 1 \end{cases} & & \\ & \xrightarrow{\quad y \quad} & \\ & \xleftarrow{\quad b' \quad} & \\ D \text{ wins if } b' = b & & \end{array}$$

Figure 12: Game G'

Our goal is to construct a PPT reduction R that efficiently distinguishes between $\{G(U_n)\}_n$ and $\{U_{n+1}\}_n$ with non-negligible advantage, i.e., wins the game G (which is shown later along with the construction of R) with non-negligible advantage.

First, we define "hybrid functions" $G_i^\star : \{0,1\}^{n+1} \to \{0,1\}^{l(n)}$ for every $i \in 0, 1, \dots, l(n)$ by

$$\begin{aligned} G_i^\star(s) &\triangleq b_1 \dots b_{l(n)}, \text{ where} \\ b_1, \dots, b_{i-1} &\leftarrow \{0,1\} \\ \underbrace{x_j}_{n \text{ bits}} \,\|\, \underbrace{b_j}_{1 \text{ bit}} &:= \begin{cases} s, & j = i \\ G(x_{j-1}), & j = i+1, \dots, l(n) \end{cases}. \end{aligned}$$

and the "hybrid distributions" Hyb_i (all of which are $l(n)$ bits long) for every $i \in 0, 1, \dots, l(n)$ by

$$\mathsf{Hyb}_i \triangleq G_i^\star(U_{n+1}),$$

among which there are

$$\mathsf{Hyb}_0 = G'(U_n) \text{ and } \mathsf{Hyb}_{l(n)} = U_{l(n)}. \tag{6.2}$$

By the **hybrid lemma**, since

$$\left|\Pr[D(G'(U_n)) = 1] - \Pr[D(U_{l(n)}) = 1]\right| = \left|\Pr[D(\mathsf{Hyb}_0) = 1] - \Pr[D(\mathsf{Hyb}_{l(n)}) = 1]\right| = \varepsilon(n),$$

where D is a PPT distinguisher, then there exists some $i^* \in \{0, 1, \ldots, l(n) - 1\}$ such that

$$\left|\Pr[D(\mathsf{Hyb}_{i^*}) = 1] - \Pr[D(\mathsf{Hyb}_{i^*+1}) = 1]\right| \geq \frac{\varepsilon(n)}{l(n)},$$

that is, D efficiently distinguishes $\{\mathsf{Hyb}_{i^*}\}_n$ and $\{\mathsf{Hyb}_{i^*+1}\}_n$ with advantage at least $\frac{\varepsilon(n)}{l(n)}$. This means that out of all "hybrid games" $\mathsf{G}_i^\bigstar$ defined as follows, there is some game $\mathsf{G}_{i^*}^\bigstar$ which D wins with probability at least $\frac{1}{2} + \frac{\varepsilon(n)}{2l(n)}$.

$$\begin{array}{ccc}
\textsf{Ch} & & D \\
b \leftarrow \{0,1\} & & \\
y \leftarrow \begin{cases} \mathsf{Hyb}_{i^*}, & b = 0 \\ \mathsf{Hyb}_{i^*+1}, & b = 1 \end{cases} & & \\
& \xrightarrow{\quad y \quad} & \\
& \xleftarrow{\quad b' \quad} & \\
& D \text{ wins if } b' = b &
\end{array}$$

Figure 13: Game $\mathsf{G}_i^\bigstar$, for each $i \in \{0, 1, \ldots, l(n) - 1\}$

Knowing this, we therefore construct the reduction R as described in the following game G:

$$\begin{array}{ccccc}
\textsf{Ch} & & R & & D \\
b \leftarrow \{0,1\} & & & & \\
s \leftarrow \begin{cases} G(U_n), & b = 0 \\ U_{n+1}, & b = 1 \end{cases} & & & & \\
& \xrightarrow{\quad s \quad} & & & \\
& & i \xleftarrow{\$} \{0, 1, \ldots, l(n) - 1\} & & \\
& & & \xrightarrow{\quad G_{i+1}^\bigstar(s) \quad} & \\
& & & \xleftarrow{\quad b' \quad} & \\
& \xleftarrow{\quad b' \quad} & & & \\
& & R \text{ wins if } b' = b & &
\end{array}$$

Figure 14: Game G

wherein by sending $G_{i+1}^\bigstar(s)$ we mean that R samples a string $y \in \{0,1\}^{l(n)}$ from the distribution $G_{i+1}^\bigstar(s)$ by computing

$$y := b_1 \ldots b_{l(n)}, \text{ where}$$
$$b_1, \ldots, b_{i-1} \leftarrow \{0, 1\}$$
$$\underbrace{x_j}_{n \text{ bits}} \| \underbrace{b_j}_{1 \text{ bit}} := \begin{cases} s, & j = i \\ G(x_{j-1}), & j = i+1, \ldots, l(n) \end{cases},$$

and then sends y to D.

The advantage of R in distinguishing between $\{G(U_n)\}_n$ and $\{U_{n+1}\}_n$ is

$$
\begin{aligned}
& \left|\Pr[R(G(U_n)) = 1] - \Pr[R(U_{n+1}) = 1]\right| \\
= & \left| \Pr_{i \leftarrow \{0,\dots,l(n)-1\}} \left[D(G^{\bigstar}_{i+1}(G(U_n))) = 1\right] - \Pr_{i \leftarrow \{0,\dots,l(n)-1\}} \left[D(G^{\bigstar}_{i+1}(U_{n+1})) = 1\right] \right| && \text{(by def. of } R\text{)} \\
\overset{(🍄)}{=} & \left| \Pr_{i \leftarrow \{0,\dots,l(n)-1\}} \left[D(G^{\bigstar}_{i}(U_{n+1})) = 1\right] - \Pr_{i \leftarrow \{0,\dots,l(n)-1\}} \left[D(G^{\bigstar}_{i+1}(U_{n+1})) = 1\right] \right| \\
= & \left| \Pr_{i \leftarrow \{0,\dots,l(n)-1\}} [D(\mathsf{Hyb}_i) = 1] - \Pr_{i \leftarrow \{0,\dots,l(n)-1\}} \left[D(\mathsf{Hyb}_{i+1}) = 1\right] \right| && \text{(by def. of } \mathsf{Hyb}_i\text{)} \\
= & \left| \sum_{i'=0}^{l(n)-1} \Pr_{i \leftarrow \{0,\dots,l(n)-1\}} [i = i'] \Pr\left[D(\mathsf{Hyb}_{i'}) = 1\right] - \sum_{i'=0}^{l(n)-1} \Pr_{i \leftarrow \{0,\dots,l(n)-1\}} [i = i'] \Pr\left[D(\mathsf{Hyb}_{i'+1}) = 1\right] \right| && \text{(by cond. prob.)} \\
= & \left| \sum_{i=0}^{l(n)-1} \frac{1}{l(n)} \cdot \Pr\left[D(\mathsf{Hyb}_i) = 1\right] - \sum_{i=0}^{l(n)-1} \frac{1}{l(n)} \cdot \Pr\left[D(\mathsf{Hyb}_{i+1}) = 1\right] \right| \\
= & \frac{1}{l(n)} \left| \sum_{i=0}^{l(n)-1} \left(\Pr\left[D(\mathsf{Hyb}_i) = 1\right] - \Pr\left[D(\mathsf{Hyb}_{i+1}) = 1\right] \right) \right| \\
= & \frac{1}{l(n)} \left| \Pr\left[D(\mathsf{Hyb}_0) = 1\right] - \Pr\left[D(\mathsf{Hyb}_{l(n)}) = 1\right] \right| && \text{(by telescoping sum)} \\
= & \frac{1}{l(n)} \left| \Pr\left[D(G'(U_n)) = 1\right] - \Pr\left[D(U_{l(n)}) = 1\right] \right| && \text{(by (6.2))} \\
= & \frac{1}{l(n)} \cdot \varepsilon(n), && \text{(by (6.1))}
\end{aligned}
$$

which is non-negligible because $\varepsilon(n)$ is non-negligible and $l(n) = \mathsf{poly}(n)$. Notice that the step (🍄) above follows from a simple observation:

$$G^{\bigstar}_{i+1}(G(U_n)) = G^{\bigstar}_{i}(U_{n+1}) \quad \text{for all } i \in \{0, 1, \dots, l(n) - 1\},$$

which is given directly by the definition of $G^{\bigstar}_i$.

On the other hand, the R clearly runs in PPT by construction because D also runs in PPT.

Hence, the existence of the PPT R, which distinguishes between $\{G(U_n)\}_n$ and $\{U_{n+1}\}_n$ with non-negligible advantage, by definition breaks the pseudorandomness of $\{G(U_n)\}_n$. This implies G is not a PRG, contradicting to the assumption that G is. This completes our proof, yay. ⊛

Problem 7 (Complete the Goldreich-Levin).

Recall from the proof of the Goldreich-Levin Theorem that we can only assume that there exists a PPT adversary $\mathbf{Adv}^H$ such that

$$\Pr_{x \leftarrow \{0,1\}^*}\left[\mathbf{Adv}^H(1^{2n}, g(x,r)) = h(x,r)\right] \geq \frac{1}{2} + \varepsilon,$$

and we construct the reduction $\mathcal{R}$ as follows: for input $(y = f(x), r)$,

1. let $m = \mathsf{poly}(\frac{1}{\varepsilon}) = \frac{n}{\varepsilon^2}$ and $k = \log m$;
2. samples $r_1, \ldots, r_k \leftarrow \{0,1\}^n$ and guess b_j (hope for $b_j = \langle x, r_j \rangle$) for each $j \in [k]$;
3. for each $I \subseteq [k]$, set $r_I = \bigoplus_{j \in I} r_j$, $b_I = \bigoplus_{j \in I} b_j$ and
4. for each $i \in [n]$, set $r_{iI} = r_I \oplus e_i$, where e_i is the i-th standard unit vector;
5. independently send (y, r_{iI}) to $\mathbf{Adv}^H$ and $\mathbf{Adv}^H$ returns b'_{iI};
6. let $c_{iI} = b_I \oplus b'_{iI}$ and set $x'_i = \mathsf{majority}\{c_{iI}\}_{I \neq \varnothing}$;
7. outputs $x' = (x'_1, \ldots, x'_n)$ together with the input r.

(i) Show that for each $i \in [n]$, $\{r_{iI}\}_{I \neq \varnothing}$ is pairwise independent.

(ii) Deduce from (i) that, conditioning on $b_j = \langle x, r_j \rangle$ for all $j \in [k]$ (i.e. all guesses are correct), then for each $i \in [n]$, $\{c_{iI}\}_{i \neq \varnothing}$ is pairwise independent.

(iii) Finally prove the following inequality by Chebychev inequality

$$\Pr\left[x'_i \neq x_i \,\middle|\, x \in G_1 \wedge \bigwedge_{j \in [k]} b_j = \langle x, r_j \rangle\right] \leq \frac{1}{\varepsilon^2 m},$$

where

$$G_1 \triangleq \left\{x \in \{0,1\}^n \,\middle|\, \Pr_{r \leftarrow \{0,1\}^n}\left[\mathbf{Adv}^H(1^{2n}, (f(x), r)) = h(x,r)\right] > \frac{1}{2} + \frac{\varepsilon}{2}\right\}.$$

Theorem (Chebyshev's Inequality). Let $X_1, \ldots, X_m$ be pairwise independent random variables with the same expectation μ and variance σ^2. Then for every $\delta > 0$,

$$\Pr\left[\left|\frac{1}{m}\sum_{i=1}^m X_i - \mu\right| \geq \delta\right] \leq \frac{\sigma^2}{\delta^2 m}.$$

Answer (i). By definition, our goal is to prove that r_{iI_1} and r_{iI_2} are independent for every distinct $I_1, I_2 \in \mathcal{P}([k]) \setminus \{\varnothing\}$ and each $i \in [n]$.

Let $i \in [n]$ and $I_1, I_2 \in \mathcal{P}([k]) \setminus \{\varnothing\}$ be distinct sets. Notice that

$$r_{iI_1} \perp\!\!\!\perp r_{iI_2} \overset{\text{def}}{\iff} r_{I_1} \oplus e_i \perp\!\!\!\perp r_{I_2} \oplus e_i \impliedby r_{I_1} \perp\!\!\!\perp r_{I_2} \overset{\text{def}}{\iff} \bigoplus_{j \in I_1} r_j \perp\!\!\!\perp \bigoplus_{j \in I_2} r_j,$$

so it suffices to show $\bigoplus_{j \in I_1} r_j$ and $\bigoplus_{j \in I_2} r_j$ are independent.

Goal.
$\Pr\left[\bigoplus_{j \in I_1} r_j = x \wedge \bigoplus_{j \in I_2} r_j = y\right] = \Pr\left[\bigoplus_{j \in I_1} r_j = x\right] \Pr\left[\bigoplus_{j \in I_2} r_j = y\right]$ for all $x, y \in \{0,1\}^n$.

To see this, notice that since $I_1 \neq I_2$, $I_1 \triangle I_2 \neq \varnothing$, allowing us to pick an arbitrary $i^* \in I_1 \setminus I_2 \equiv (I_1 \setminus I_2) \cup (I_2 \setminus I_1)$. WLOG, assume that $I_1 \setminus I_2 \neq \varnothing$. Next, observe that for any $x, y \in \{0,1\}^n$,

$$
\begin{aligned}
\Pr\left[\bigoplus_{j\in I_1} r_j = x \wedge \bigoplus_{j\in I_2} r_j = y\right] &= \Pr\left[\left(\bigoplus_{j\in I_1\setminus\{i^*\}} r_j\right) \oplus r_{i^*} = x \wedge \bigoplus_{j\in I_2} r_j = y\right] \\
&= \Pr\left[\bigvee_{z\in\{0,1\}^n}\left(\bigoplus_{j\in I_1\setminus\{i^*\}} r_j = z \wedge r_{i^*} = z\oplus x\right) \wedge \bigoplus_{j\in I_2} r_j = y\right] \\
&= \Pr\left[\bigvee_{z\in\{0,1\}^n}\left(\bigoplus_{j\in I_1\setminus\{i^*\}} r_j = z \wedge r_{i^*} = z\oplus x \wedge \bigoplus_{j\in I_2} r_j = y\right)\right] \\
&= \sum_{z\in\{0,1\}^n} \Pr\left[\bigoplus_{j\in I_1\setminus\{i^*\}} r_j = z \wedge r_{i^*} = z\oplus x \wedge \bigoplus_{j\in I_2} r_j = y\right] \qquad \text{(by disjointness)} \\
&= \sum_{z\in\{0,1\}^n} \Pr\left[r_{i^*} = z\oplus x\right] \Pr\left[\bigoplus_{j\in I_1\setminus\{i^*\}} r_j = z \wedge \bigoplus_{j\in I_2} r_j = y\right] \\
&\qquad \text{(Since } i^* \notin (I_1\setminus\{i^*\}) \cup I_2 \text{ and } r_1,\ldots,r_k \text{ are mutually independent,} \\
&\qquad\ r_{i^*} \text{ is independent of } \left(\textstyle\bigoplus_{j\in I_1\setminus\{i^*\}} r_j, \bigoplus_{j\in I_2} r_j\right).\text{)} \\
&= \frac{1}{2^n}\sum_{z\in\{0,1\}^n} \Pr\left[\bigoplus_{j\in I_1\setminus\{i^*\}} r_j = z \wedge \bigoplus_{j\in I_2} r_j = y\right] \qquad \text{(since } r_{i^*} \text{ is uniformly distributed)} \\
&= \frac{1}{2^n} \Pr\left[\bigoplus_{j\in I_2} r_j = y\right] \qquad \text{(by marginal probability)} \\
&= \Pr\left[\bigoplus_{j\in I_1} r_j = x\right] \Pr\left[\bigoplus_{j\in I_2} r_j = y\right], \\
&\qquad \text{(since } \textstyle\bigoplus_{j\in I_1} r_j \text{ is uniformly distributed by Observation 7.1)}
\end{aligned}
$$

where in the last step we've used the following observation.

Observation 7.1. $r_{I_1} = \bigoplus_{j\in I_1} r_j$ is uniformly distributed.

Proof. Since $I_1 \neq \varnothing$, we may pick $t \in I_1$. If $|I_1| = 1$, then $\bigoplus_{j\in I_1} r_j = r_t$, which is clearly uniformly distributed, and thus we are done. Therefore we may assume that $|I_1| > 1$.

Observe that $(\{0,1\}^n, \oplus)$ forms a group, as shown in Problem 1(i), and recall the following result proven in Problem 1(ii):

Lemma (Problem 1(ii)). Let U and V be random variables over a group $(G, \cdot)$. If U is uniformly distributed and U, V are independent, then $U \cdot V$ is also uniformly distributed.

Applying this lemma, we see that since r_t is uniformly distributed and r_t is independent of $\bigoplus_{j\in I_1\setminus\{t\}} r_j$ (which is because $r_1, \ldots, r_k$ are mutually independent),

$$\bigoplus_{j\in I_1} r_j = r_t \oplus \bigoplus_{j\in I_1\setminus\{t\}} r_j$$

is also uniformly distributed. ■

By definition, this means $r_{I_1} = \bigoplus_{j\in I_1} r_j$ and $r_{I_2} = \bigoplus_{j\in I_2} r_j$ are independent, as desired. ⊛

Answer (ii). By definition, our goal is to prove that c_{iI_1} and c_{iI_2} are independent for every distinct $I_1, I_2 \in \mathcal{P}([k]) \setminus \{\varnothing\}$ and each $i \in [n]$.

Let $i \in [n]$ and $I_1, I_2 \in \mathcal{P}([k]) \setminus \{\varnothing\}$ be distinct sets.

Goal. $c_{iI_1} \perp\!\!\!\perp c_{iI_2}$

Idea. Recall that we've already shown in subproblem (i) that, given nonempty and distinct sets I_1 and I_2, we know r_{I_1} and r_{I_2} are independent. Hence, our idea is: if we could express c_{iI_1} and c_{iI_2} as a (deterministic) function of r_{I_1} and r_{I_2} respectively (say, $c_{iI_1} = f(r_{I_1})$ and $c_{iI_2} = f(r_{I_2})$ for some f), then the following lemma would guarantee f preserves the independence of its inputs, i.e.,

$$r_{I_1} \perp\!\!\!\perp r_{I_2} \implies f(r_{I_1}) \perp\!\!\!\perp f(r_{I_2}) \iff c_{iI_1} \perp\!\!\!\perp c_{iI_2}.$$

Since LHS holds, the proof would be completed.

Lemma 7.1. Let $X_1, \dots, X_n$ be random variables. If $X_1, \dots, X_n$ are jointly independent, then so are $f_1(X_1), \dots, f_n(X_n)$ for any deterministic functions $f_1, \dots, f_n$.

Proof. For every $\alpha_1, \dots, \alpha_n$ in the supports of $f_1(X_1), \dots, f_n(X_n)$, respectively, it holds that

$$\begin{aligned}
&\Pr[f_1(X_1) = \alpha_1 \wedge \cdots \wedge f_n(X_n) = \alpha_n] \\
=& \Pr[X_1 \in f_1^{-1}(\alpha_1) \wedge \cdots \wedge X_n \in f_n^{-1}(\alpha_n)] \\
=& \Pr[X_1 \in f_1^{-1}(\alpha_1)] \cdots \Pr[X_n \in f_n^{-1}(\alpha_n)] \qquad \text{(since } X_1, \dots, X_n \text{ are jointly indep.)} \\
=& \Pr[f_1(X_1) = \alpha_1] \cdots \Pr[f_n(X_n) = \alpha_n].
\end{aligned}$$

■

Corollary 7.1. Let X and Y be random variables. If X and Y are independent, then so are $f(X)$ and $g(Y)$ for any deterministic functions f, g.

Wonderful as the strategy sounds, we couldn't really find such deterministic function f since $\mathbf{Adv}^H$ is probabilistic. However, it turns out that we can find an f such that $c_{iI_1} = f(r_{I_1}, \vec{r_1})$ and $c_{iI_2} = f(r_{I_2}, \vec{r_2})$, where $\vec{r_1}, \vec{r_2}$ are random variables "extracted" from $\mathbf{Adv}^H$.

In order to leverage Corollary 7.1 to show that the **probabilistic (randomized)** algorithm $\mathbf{Adv}^H$ preserves independence of inputs, we may "transform" $\mathbf{Adv}^H$ into a deterministic algorithm (which may be viewed as a function) $\mathbf{Adv}_D^H$ by explicitly specifying by $\vec{r}$ the internal random vector it samples:

$$\mathbf{Adv}^H(\,\cdot\,) \equiv \mathbf{Adv}_D^H(\,\cdot\,; \vec{r}) \quad \text{where } \vec{r} \leftarrow \{0,1\}^{\ell(n)}. \tag{7.1}$$

Having a deterministic algorithm $\mathbf{Adv}_D^H$, we can finally rewrite c_{iI_1} (and c_{iI_2} as well) as

$$
\begin{aligned}
c_{iI_1} &= b_{I_1} \oplus b'_{iI_1} && \text{(by def.)}\\
&= \left(\bigoplus_{j\in I} b_j\right) \oplus \mathbf{Adv}^H(y, r_{I_1} \oplus e_i) && \text{(by def.)}\\
&= \left(\bigoplus_{j\in I_1} \langle x, r_j\rangle\right) \oplus \mathbf{Adv}^H(y, r_{I_1} \oplus e_i) && \text{(by def.)}\\
&= \left(\bigoplus_{j\in I_1} \bigoplus_{t=1}^{n} x_t (r_j)_t\right) \oplus \mathbf{Adv}^H(y, r_{I_1} \oplus e_i) && \text{(by def.)}\\
&= \left(\bigoplus_{t=1}^{n} \bigoplus_{j\in I_1} x_t (r_j)_t\right) \oplus \mathbf{Adv}^H(y, r_{I_1} \oplus e_i) &&\\
&= \left(\bigoplus_{t=1}^{n} x_t \left(\bigoplus_{j\in I_1} r_j\right)_t\right) \oplus \mathbf{Adv}^H(y, r_{I_1} \oplus e_i) &&\\
&= \left(\bigoplus_{t=1}^{n} x_t (r_{I_1})_t\right) \oplus \mathbf{Adv}^H(y, r_{I_1} \oplus e_i) && \text{(by def. of } r_{I_1})\\
&= \langle x, r_{I_1}\rangle \oplus \mathbf{Adv}^H(y, r_{I_1} \oplus e_i) && \text{(by def. of } \langle\,\cdot\,,\,\cdot\,\rangle)\\
&= \langle x, r_{I_1}\rangle \oplus \mathbf{Adv}_D^H(y, r_{I_1} \oplus e_i; \vec{r_1}) && \text{(by the transformation (7.1))}
\end{aligned}
$$

where $\vec{r_1}$ is a fresh random vector sampled from some distribution. Likewise, we rewrite $c_{iI_2} = \langle x, r_{I_2}\rangle \oplus \mathbf{Adv}_D^H(y, r_{I_2} \oplus e_i; \vec{r_1})$ where $\vec{r_2}$ is a fresh random vector $\vec{r_2}$ independent of $\vec{r_1}$ and sampled from some distribution. For simplicity, let $f : \{0,1\}^n \times \{0,1\}^{\ell(n)} \to \{0,1\}$ be defined by

$$f(s, \vec{r}) = \langle x, s\rangle \oplus \mathbf{Adv}_D^H(y, s \oplus e_i; \vec{r}),$$

where $\ell(n) = |\vec{r_1}| = |\vec{r_2}|$. Note that f is a deterministic function since x, y, e_i are constants and $\mathbf{Adv}_D^H$ is a deterministic algorithm. We hence have

$$c_{iI_1} = f(r_{I_1}, \vec{r_1}), \quad c_{iI_2} = f(r_{I_2}, \vec{r_2}) \tag{7.2}$$

for some fresh random vectors $\vec{r_1}, \vec{r_2}$.

> Goal. $f(r_{I_1}, \vec{r_1}) \perp\!\!\!\perp f(r_{I_2}, \vec{r_2})$

Last but not least, notice that since (1) $r_{I_1} \perp\!\!\!\perp r_{I_2}$ (proven in subproblem (i)), (2) $\vec{r_1} \perp\!\!\!\perp \vec{r_2}$, and (3) $(\vec{r_1}, \vec{r_2}) \perp\!\!\!\perp (r_{I_1}, r_{I_2})$, by an additional claim Lemma 7.2 (whose proof will be given later), we know that $(r_{I_1}, \vec{r_1}) \perp\!\!\!\perp (r_{I_2}, \vec{r_2})$, both of which are exactly inputs to f.

Putting everything together, as f is a deterministic function, applying Corollary 7.1 we obtain

$$f(r_{I_1}, \vec{r_1}) \perp\!\!\!\perp f(r_{I_2}, \vec{r_2})$$

as the inputs $(r_{I_1}, \vec{r_1})$ and $(r_{I_2}, \vec{r_2})$ are independent. Equivalently by (7.2), we have

$$c_{iI_1} \perp\!\!\!\perp c_{iI_2},$$

as desired.

Lemma 7.2. Let X_1, X_2, Y_1, Y_2 be random variables. If $X_1 \perp\!\!\!\perp Y_1$, $X_2 \perp\!\!\!\perp Y_2$, and $(X_1, Y_1) \perp\!\!\!\perp (X_2, Y_2)$, then $(X_1, X_2) \perp\!\!\!\perp (Y_1, Y_2)$.

Proof of lemma. For any a, b, c, d,

$$\begin{aligned}
p_{(X_1,X_2),(Y_1,Y_2)}((a,b),(c,d)) &= p_{X_1,X_2,Y_1,Y_2}(a,b,c,d) \\
&= p_{(X_1,Y_1),(X_2,Y_2)}((a,c),(b,d)) \\
&= p_{X_1,Y_1}(a,c)p_{X_2,Y_2}(b,d) && (\because (X_1,Y_1) \perp\!\!\!\perp (X_2,Y_2)) \\
&= p_{X_1}(a)p_{Y_1}(c)p_{X_2}(b)p_{Y_2}(d) && (\because X_1 \perp\!\!\!\perp Y_1 \wedge X_2 \perp\!\!\!\perp Y_2) \\
&= p_{X_1}(a)p_{X_2}(b)p_{Y_1}(c)p_{Y_2}(d) \\
&= p_{X_1,X_2}(a,b)p_{Y_1,Y_2}(c,d). && (\because (X_1,Y_1) \perp\!\!\!\perp (X_2,Y_2) \implies X_1 \perp\!\!\!\perp X_2 \wedge Y_1 \perp\!\!\!\perp Y_2)
\end{aligned}$$

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Answer (iii). For each $i \in [n]$ and each $I \in \mathcal{P}([k]) \setminus \{\varnothing\}$, define random variables

$$Z_{iI} \triangleq \begin{cases} 1, & \text{if } c_{iI} = x_i \\ 0, & \text{otherwise} \end{cases}.$$

Since $\{Z_{iI}\}_{I \neq \varnothing}$ are functions of $\{c_{iI}\}_{I \neq \varnothing}$, which are proven to be pairwise independent in subproblem (ii), by Corollary 7.1, $\{Z_{iI}\}_{I \neq \varnothing}$ are also pairwise independent. Moreover, for any $i \in [n]$ and $I \in \mathcal{P}([k]) \setminus \{\varnothing\}$, the conditional expectation μ_{Z_i} and conditional variance $\sigma^2_{Z_i}$ of Z_{iI} given $x \in G_1$ (i.e. x is good) and $b_j = \langle x, r_j \rangle \ \forall j \in [k]$ (i.e. all guesses are correct) are the same:

$$\begin{aligned}
\mu_{Z_i} &\triangleq \mathbb{E}\left[Z_{iI} \,\middle|\, x \in G_1 \wedge \bigwedge_{j \in [k]} b_j = \langle x, r_j \rangle \right] \\
&= \Pr_{r_1,\ldots,r_k \leftarrow \{0,1\}^n} \left[c_{iI} = x_i \,\middle|\, x \in G_1 \wedge \bigwedge_{j \in [k]} b_j = \langle x, r_j \rangle \right] \\
&\overset{(🍄)}{\geq} \Pr_{r_1,\ldots,r_k \leftarrow \{0,1\}^n} \left[\mathbf{Adv}^H(f(x), r_I \oplus e_i) = h(x, r_I \oplus e_i) \,\middle|\, x \in G_1 \wedge \bigwedge_{j \in [k]} b_j = \langle x, r_j \rangle \right] \\
&= \Pr_{r, r_1,\ldots,r_k \leftarrow \{0,1\}^n} \left[\mathbf{Adv}^H(f(x), r) = h(x, r) \,\middle|\, x \in G_1 \wedge \bigwedge_{j \in [k]} b_j = \langle x, r_j \rangle \right] \\
&\qquad \text{(since } r_I \text{ is uniformly distributed by Observation 7.1 and so is } r_I \oplus e_i) \\
&= \Pr_{r \leftarrow \{0,1\}^n} \left[\mathbf{Adv}^H(f(x), r) = h(x, r) \,\middle|\, x \in G_1 \right] \qquad \text{(by independence)} \\
&> \frac{1}{2} + \frac{\varepsilon(n)}{2} \qquad \text{(by definition of } G_1) \\
\sigma^2_{Z_i} &\triangleq \mathrm{Var}\left[Z_{iI} \,\middle|\, x \in G_1 \wedge \bigwedge_{j \in [k]} b_j = \langle x, r_j \rangle \right] \\
&= \mu_{Z_i}(1 - \mu_{Z_i}),
\end{aligned}$$

where the correctness of the step (🍄) is proven in class. Note that each Z_{iI} has indeed the same expectation and thus variance since we've eliminated I in the derivation of μ_{Z_i} before any inequalities.

Now we deduce our final result:

$$
\begin{aligned}
&\Pr_{r_1,\ldots,r_k\leftarrow\{0,1\}^n}\left[x_i' \neq x_i \;\middle|\; x \in G_1 \wedge \bigwedge_{j\in[k]} b_j = \langle x, r_j\rangle\right] \\
=&\Pr_{r_1,\ldots,r_k\leftarrow\{0,1\}^n}\left[\text{majority}\{c_{iI}\}_{I\neq\varnothing} \neq x_i \;\middle|\; x \in G_1 \wedge \bigwedge_{j\in[k]} b_j = \langle x, r_j\rangle\right] \\
=&\Pr_{r_1,\ldots,r_k\leftarrow\{0,1\}^n}\left[\frac{1}{m}\sum_{I\in\mathcal{P}([k])\setminus\{\varnothing\}} Z_{iI} \leq \frac{1}{2} \;\middle|\; x \in G_1 \wedge \bigwedge_{j\in[k]} b_j = \langle x, r_j\rangle\right] \\
\leq&\Pr_{r_1,\ldots,r_k\leftarrow\{0,1\}^n}\left[\frac{1}{m}\sum_{I\in\mathcal{P}([k])\setminus\{\varnothing\}} Z_{iI} \leq \mu_{Z_i} - \frac{\varepsilon(n)}{2} \;\middle|\; x \in G_1 \wedge \bigwedge_{j\in[k]} b_j = \langle x, r_j\rangle\right] \qquad \left(\text{since } \mu_{Z_i} > \frac{1}{2} + \frac{\varepsilon(n)}{2}\right) \\
\leq&\Pr_{r_1,\ldots,r_k\leftarrow\{0,1\}^n}\left[\left|\frac{1}{m}\sum_{I\in\mathcal{P}([k])\setminus\{\varnothing\}} Z_{iI} - \mu_{Z_i}\right| \geq \frac{\varepsilon(n)}{2} \;\middle|\; x \in G_1 \wedge \bigwedge_{j\in[k]} b_j = \langle x, r_j\rangle\right] \\
\leq&\frac{\sigma^2_{Z_i}}{\left(\frac{\varepsilon(n)}{2}\right)^2 (m)} \qquad \left(\text{by Chebyshev's inequality (conditioned on } x \in G_1 \wedge \textstyle\bigwedge_{j\in[k]} b_j = \langle x, r_j\rangle)\right) \\
=&\frac{4\mu_{Z_i}(1-\mu_{Z_i})}{\varepsilon^2(n)m} \qquad \left(\text{since } \sigma^2_{Z_i} = \mu_{Z_i}(1-\mu_{Z_i})\right) \\
\leq&\frac{1}{\varepsilon^2(n)m}. \qquad \left(\text{by AM-GM inequality: } \mu_{Z_i}(1-\mu_{Z_i}) \leq \left(\frac{\mu_{Z_i} + (1-\mu_{Z_i})}{2}\right)^2 = \frac{1}{4}\right)
\end{aligned}
$$

Hence proved. ⊛

Reference

P1 (None)

P2 (None)

P3 (None)

P4 (None)

P5 TAMC 2026 week 3 scribe by Hung-Yu Lu

P6 (None)

P7
- TAMC 2026 week 3 scribe by Ting-chieh Chiu
- Introduction to Modern Cryptography (3rd Ed.) by Jonathan Katz and Yehuda Lindell